EPFL
Ecole polytechnique federale de Lausanne

First Annual (Intratask) Meeting in the context of the
International Bioenergy and Bioeconomy Conference – IBBCE 2025
Lisbon, Portugal, December 3rd – 5th, 2025

Industrial Energy-Related Technologies and Systems iets
An IEA Technology Collaboration Programme

Task XXIV Process Integration for Industry Decarbonization

Subtask 1 - Enabling a shared database of ex-ante models of industrial processes, decarbonization technologies and energy systems featuring sustainability metrics

IETS Task XXIV Team



Dec. 5 2025



Content

- Agenda
- Summary of bilateral meetings
- Overview of the survey’s results
- Short presentations
- Breakout sessions
- Annual Report 2025
- Next steps

Agenda

- Welcome - 2 min.
- Summary of key findings from bilateral meetings - 5 min.
- Overview of the survey's results on the model-sharing framework definition:
  - Template discussion and feedback (ontologies and shared "models" structure) - 8 min.
  - Models inventory and documentation framework (database features and deployment) - 8 min.
  - Procedures ("rules of the game", expectations/concerns and confidentiality issues) - 8 min.
- Short presentations per participant country - 60 min (6 min. each participant).
- Break - 5 min.
- Discussion in breakout sessions and conclusion - 20 min.
- Explanation of the Annual Report contribution (5-page per country incl. dissemination activities). Proposed deadline: 19th Dec. - 5 min.
- Next steps, future bilateral meetings, photos and closure - 5 min.
- Any other business (AOB) - 5 min.

EPFL Team

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Colombia


Turkey


Brazil


India


Libya


India


France


Lebanon


Switzerland
Belgium

Structure and term

Time frame: 01.01.2025 – 01.11.2027

Task managers:

- Daniel Florez-Orrego
- François Maréchal

Process Integration for Industry Decarbonization

Recent Developments

The proposal for this new Task was approved by the IETS ExCo on 20 November 2024. The Task will start early 2025.

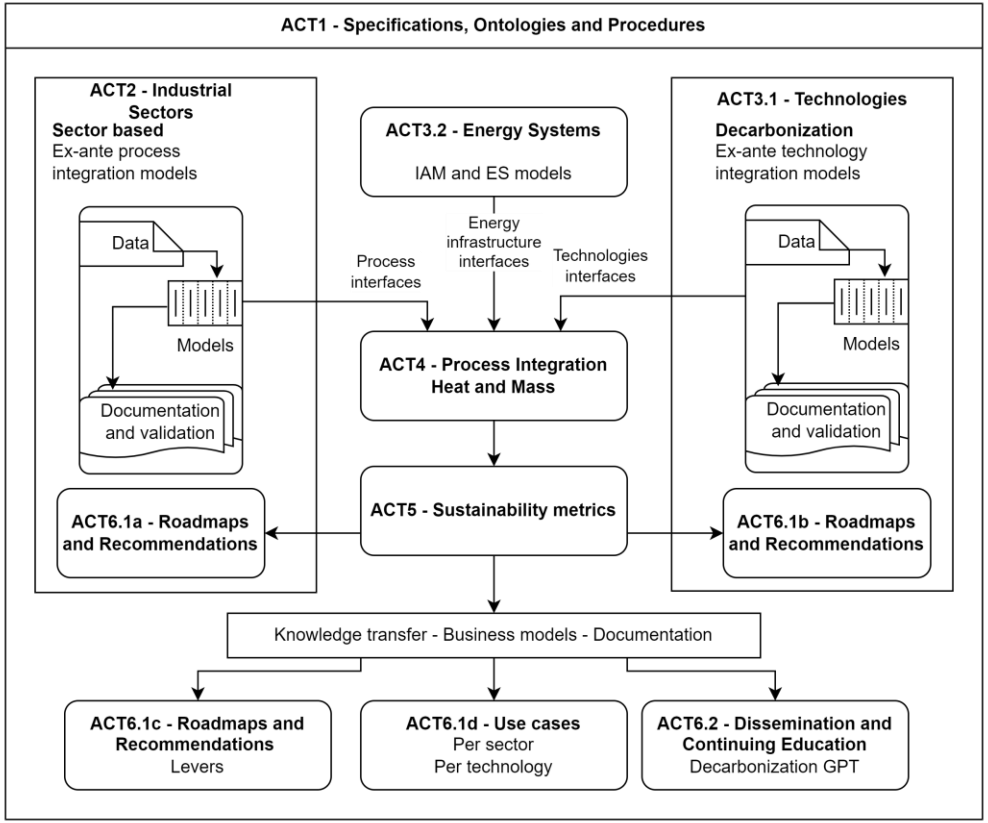
Read the Task Proposal

TASK XXIV

START DATE 2025-01-01

END DATE 2027-12-31

Read the task proposal **HERE**



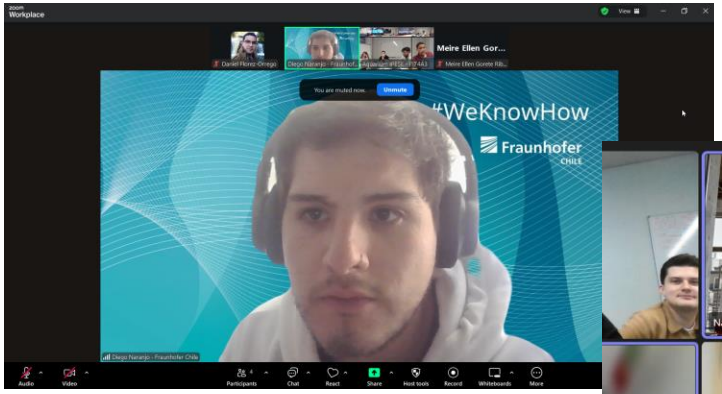
Objectives

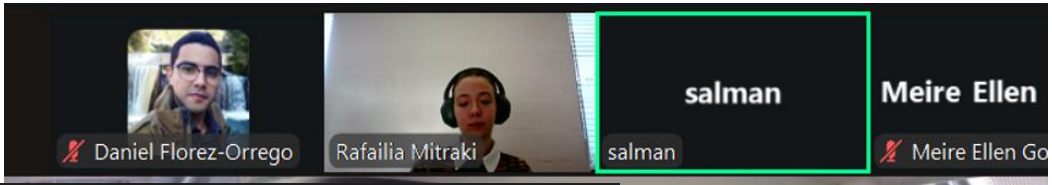
- **Define** the ontologies, specifications and procedures to generate, validate, and share models.
- **Share** knowledge and innovations in industrial processes and decarbonization technologies.
- **Discuss** impact on energy systems models, distribution networks and seasonal demands (**ETSAP**)
- **Compare** process integration methods/tools for ranking decarbonization pathways (metrics)
- **Share** experiences and case studies to outline decarbonization roadmaps and symbiosis.
- **Disseminate** through workshops, papers, scientific events, and educational programs.

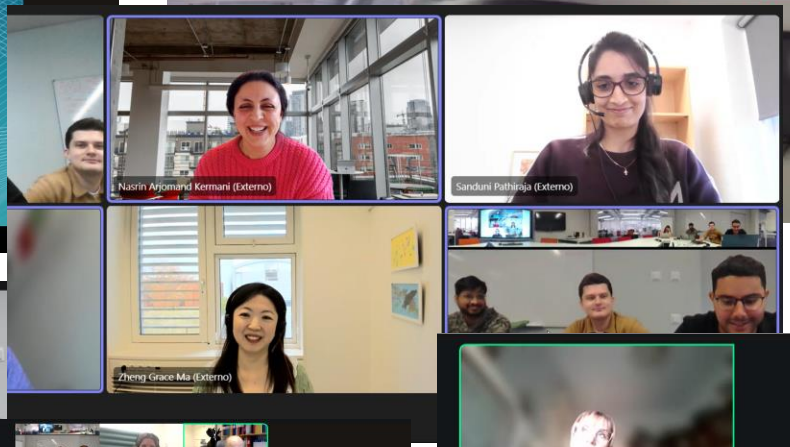


Summary of bilateral meetings

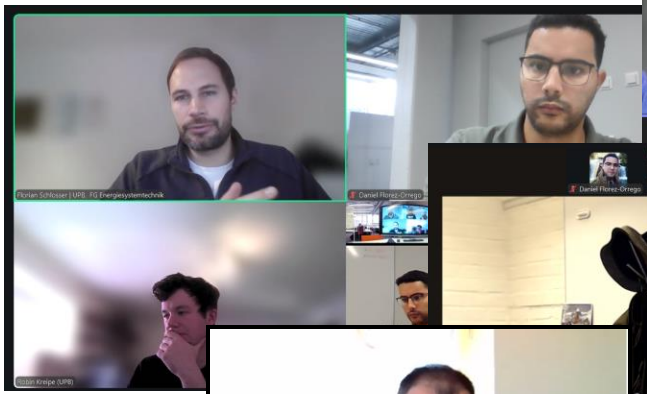
Bilateral meetings















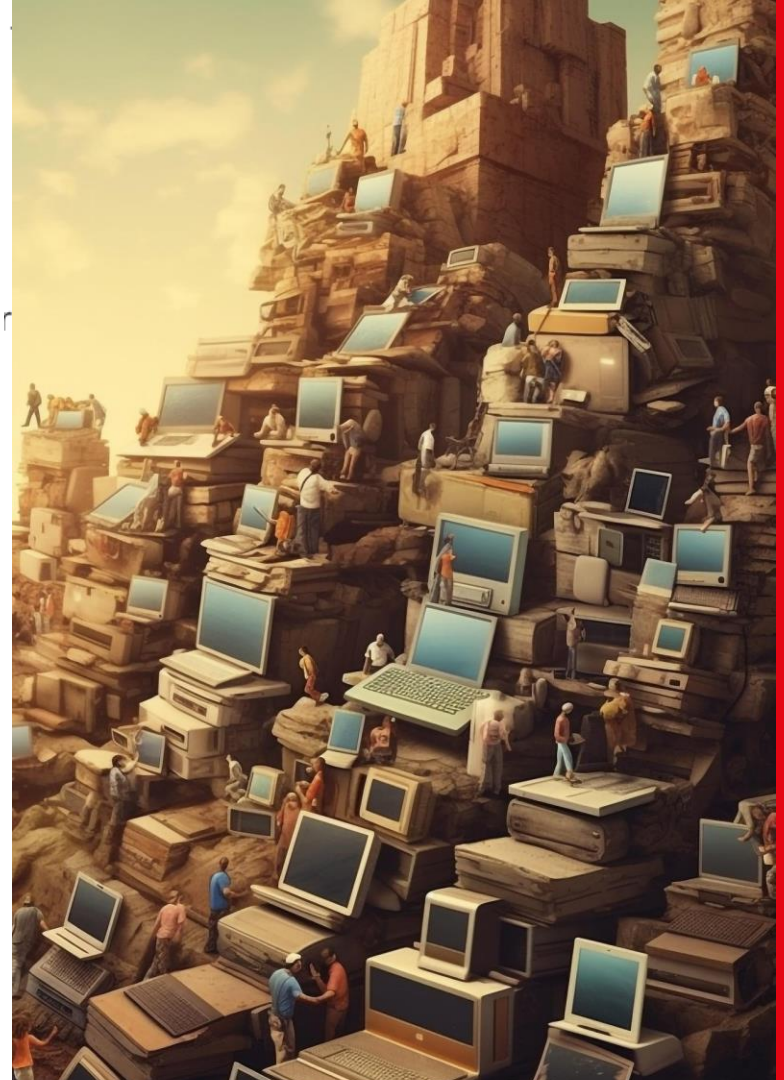


Bilateral meetings



Bilateral meetings

- **Chile**: Data exchange policies and restrictions between researchers and industrial partners
- **Belgium**: Weigh the need for different database-access interfaces for each type of stakeholders
- **Sweden**: Tackle the impact of the projects confidentiality on the models sharing between universities
- **Italy**: Assess the model portability and compatibility across scales (e.g. chemistry, engineering, control)
- **Denmark**: Define added value and costs of additional documentation compared to journal publications
- **Portugal**: Perform validation first by sub-teams of Task participants, next by the broader community
- **Germany**: Achieve data anonymization while maintaining real-world relevance
- **Switzerland**: Maximize use of open access data to promote public access to information
- **Netherlands**: Carefully evaluate open sharing options for models built on proprietary software
- **Colombia**: Promote models dissemination in broader educational platforms

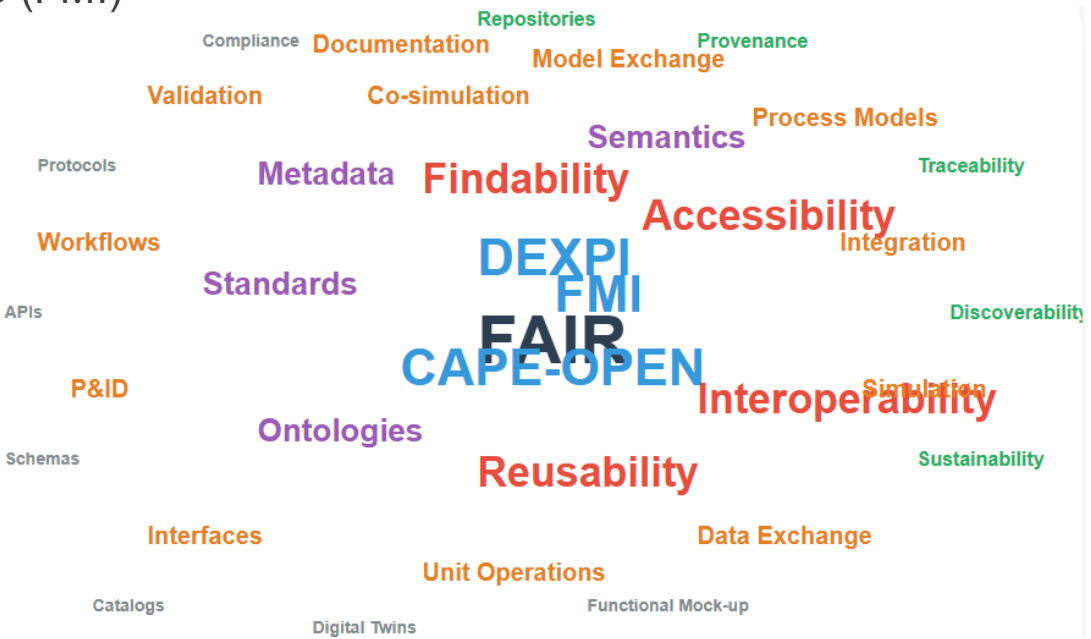


Overview of the survey results

Template discussion and feedback (ontologies and shared "models" structure)

Existing methodologies

- Model exchange standards
  - Functional mockup interface (FMI)
  - CAPE-Open
  - DEXPI
  - ...
- FAIR principles
  - Findability
  - Accessibility
  - Interoperability
  - Reusability
- Ontologies



Formal *vs* simplified frameworks

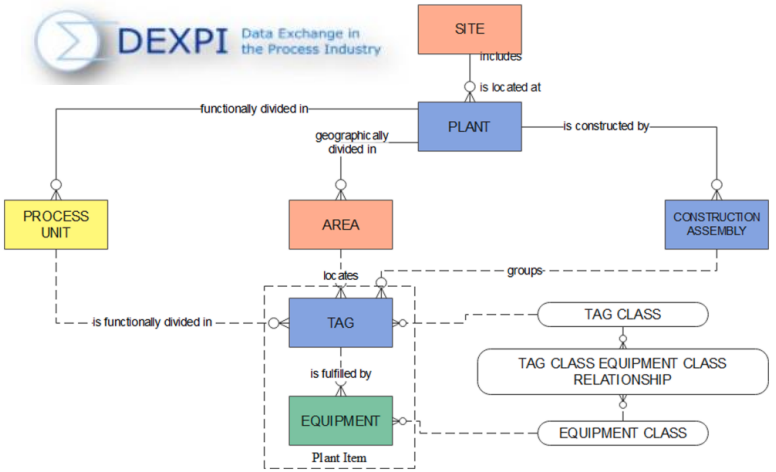
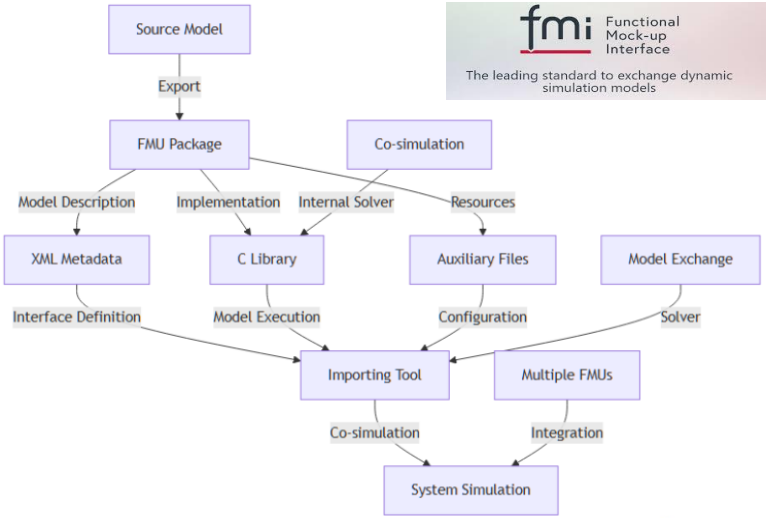


Fig. 2. Part of the CFIHOS data model (CFIHOS 2023), showing the place of TAG and EQUIPMENT objects. The objects are coloured to indicate the aspect used to classify the object: blue is a specified physical object, yellow is a functional aspect, red is a location aspect and green is a supplied physical object.

In general, the process plant industry faces significant challenges when it comes to data handover. The ISO 15926 is an international standard framework for data exchange, but the framework itself it is to unspecific to be practically usable. As a consequence, many different specializations of ISO15926 have been derived. Unfortunately, without any alignment between each other.
The DEXPI format successfully uses and extends existing standards to deliver a practically usable format. It enables workflows beyond P&ID design such as instrumentation, visualization, integration with 3D models and simulation, thus, paving the way to digitalization in the process plant industry.

https://www.sciencedirect.com/science/article/pii/S0098135423004349



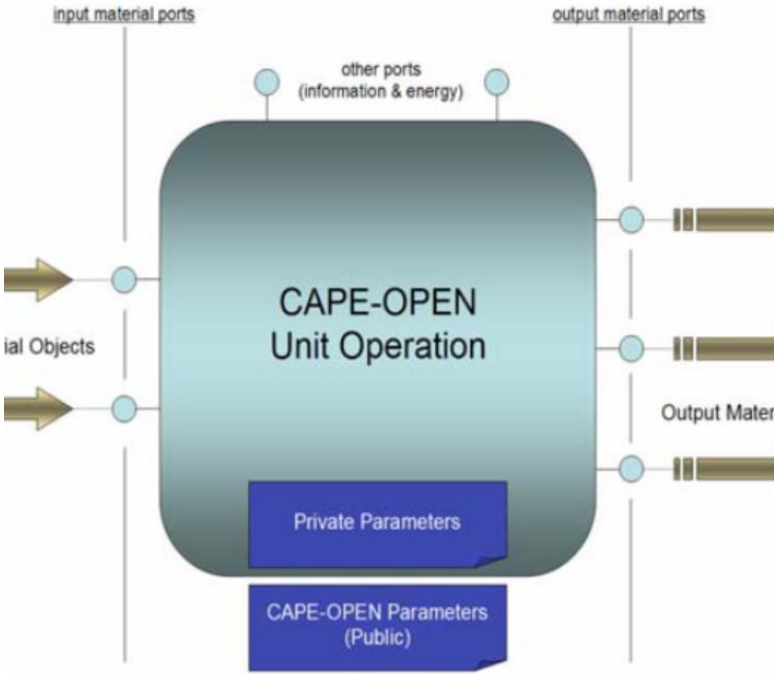
An FMU comprises several standardized components:

1. **Model Description**: XML file that defines model interfaces, parameters, and capabilities
2. **Implementation**: Compiled code (C functions) that implements the mathematical model
3. **Documentation**: Optional documentation including model descriptions and usage guidelines
4. **Resources**: Additional files including lookup tables, configuration files, and auxiliary data
5. **FMI Interface**: Standardized API that enables communication with importing simulation tools
6. **Platform Binaries**: Platform-specific compiled libraries for different operating systems

https://quix.io/glossary/functional-mock-up-unit-fmu

Open source: CAPE OPEN

- CAPE-OPEN: open standards for creating interoperable components in chemical process simulation software.
- The interface includes ports that can are material, energy and information ports.
- Parameters can be classified into private and public parameter.



https://doi.org/10.5772/14218

Model interfaces definition

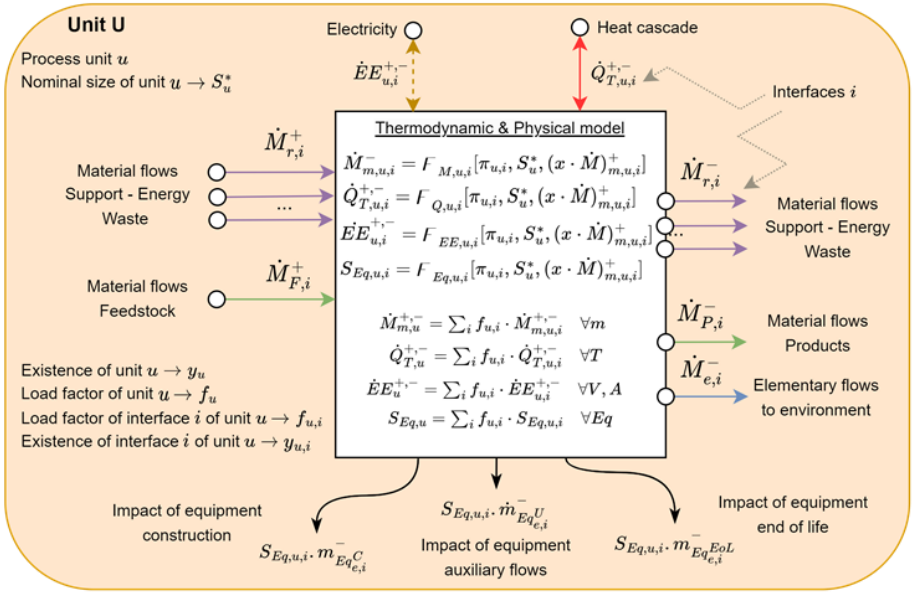
- **Metadata**: Identifier, authorship, versioning, confidentiality.
- **Context**: Connections, flows, quality, cost, impact, TRL.
- **Models**: Parameters, decision variables and assumptions.
- **Metrics**: Energy, environmental, economic indicators.

DWSIM Chemical Process Simulator

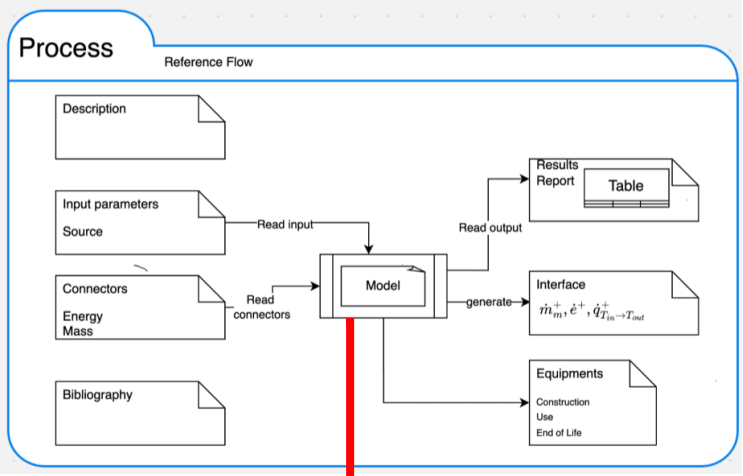
CoolProp

aspentech

Black
Grey
White



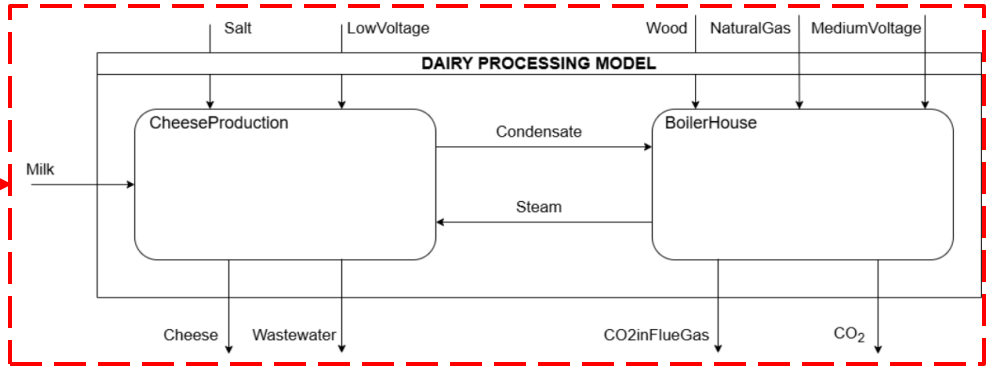
Model sharing, ontologies and procedures



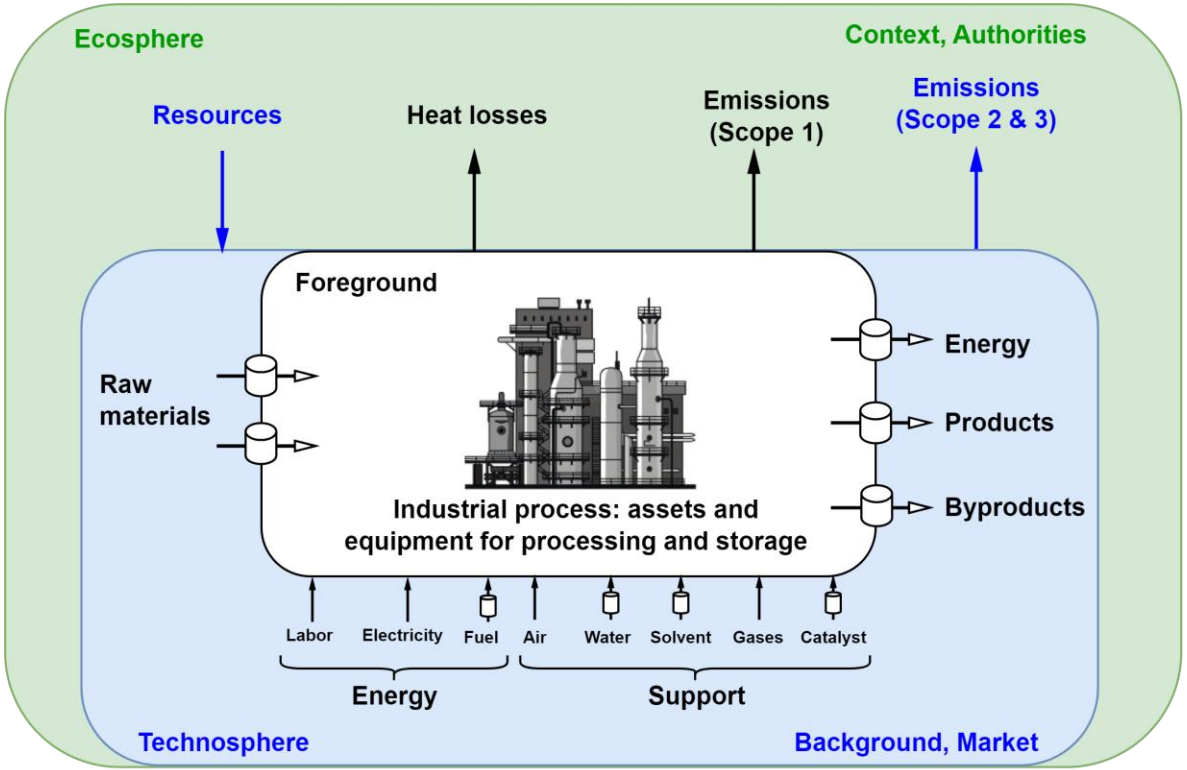
Template proposal

Model Name	DairyProcessing
Authors and Contributors	
Version	1.0
Confidentiality	Open
Creation Date	
References	
Grade	GRAY-BOX
Software	DWSIM
URL	
Last Update Date	
Keywords	Food
Reference Capacity (main product/feedstock flowrate)	10 kg/h Cheese

Metadata for univocally defining version, authorship, confidentiality, grade of detail, reference capacity, and software used.

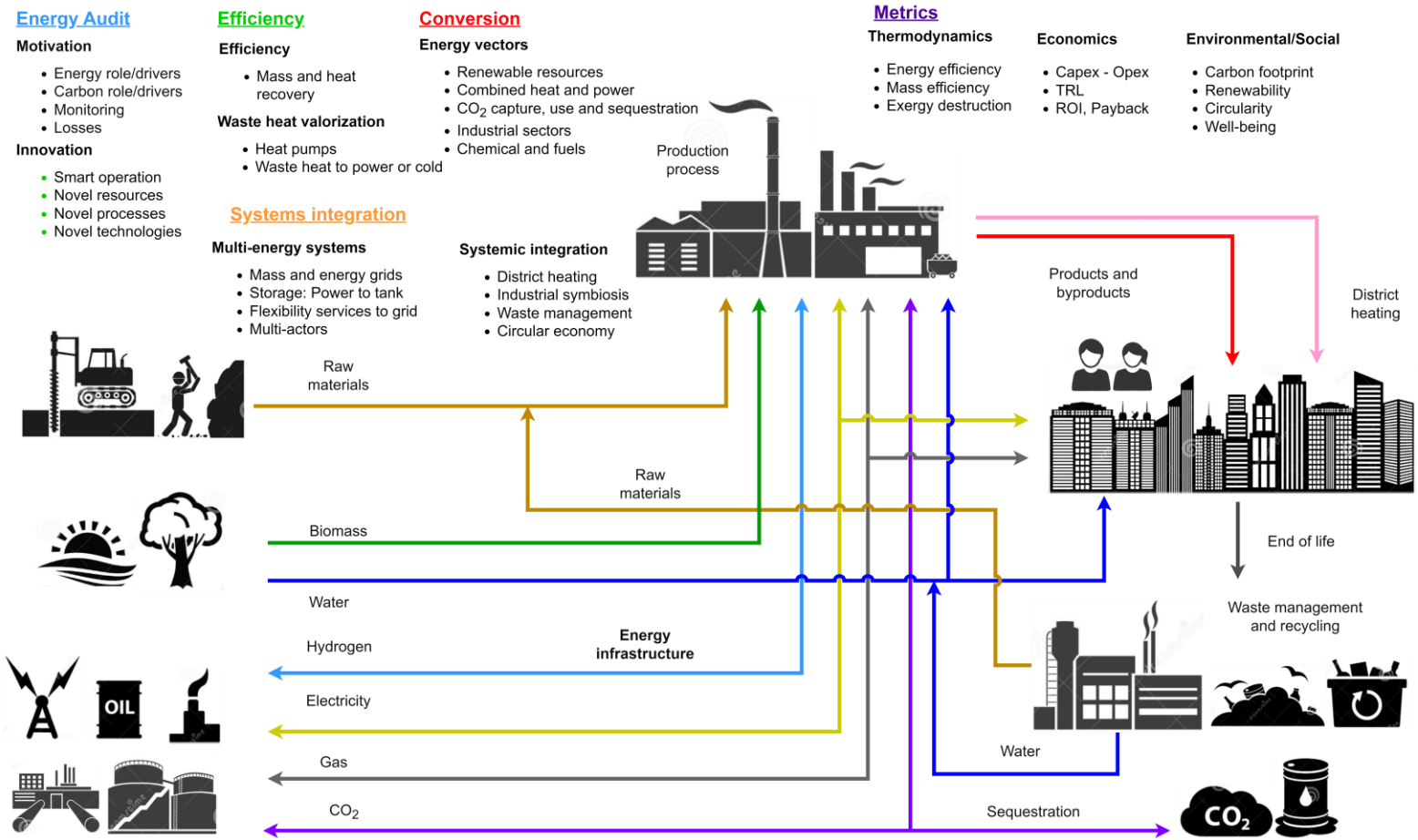


Industry, technosphere and natural resources



- Industrial process in a technosphere and an ecosphere.
- Background characteristics defines the selection of operating conditions in the foreground.
- Processes operate within a regulatory context of authorities and compliance requirements.

Engineering the decarbonization of industry



Model sharing, ontologies and procedures

EXTERNAL CONNECTORS							
Unique Identifier Number, UID	NAME (ALIAS)	TYPE	PHYSICAL UNIT	T [C]	P [BAR]	COMPOSITION [MOLE/MASS]	PHASE FRACTION [MASS]
MILK-FAT3	MILK	MASS	kg/h	25	1	MASSFRAC:FAT[0.03]&H2O[0.97]	-1 (L)
CHEESE-SOFT	CHEESE	MASS	kg/h	25	1	MASSFRAC:FAT[0.3]&H2O[0.6]&NACL[0.1]	-1 (S)
SALT-NACL	SALT	MASS	kg/h	25	1	MASSFRAC:NACL[1]	-1 (S)
WASTEWATER-SEWAGE1	WASTEWATER	MASS	kg/h	25	1	MASSFRAC:H2O[0.99]&FAT[0.01]	-1 (L)
TAP-WATER	WATER	MASS	kg/h	25	1	MASSFRAC:H2O[1]	-1 (L)
WOOD-DRY	WOOD	MASS	kg/h	25	1	MASSFRAC:C[0.47]&H[0.06]&O[0.45]&S[0]&ASH[0.02]	-1 (S)
NATURALGAS	CH4	MASS	kg/h	5	1	MASSFRAC:CH4[1]	-1 (V)
ELEC-50KV	MEDIUMVOLTAGE	ELECTRICITY	kW	-	-	-	-
ELEC-13KV	LOWVOLTAGE	ELECTRICITY	kW	-	-	-	-
FLUEGAS-M	CO2INFLUEGAS	EMISSIONS	kg/h	120	1	MOLEFRAC:CO2[0.15]&H2O[0.12]&N2[0.73]	-1 (V)
CO2-PURE	CO2	EMISSIONS	kg/h	25	1	MASSFRAC:CO2[1]	-1 (V)

Interfacing the specific models with the global database of connectors.

Engineering variables must be physically well-defined, assumed or calculated (software, formula, etc.), as well as referenced.

VARIABLES												
Name	Type	Full name	User grade	Default Value / Formula	Physical unit	Min bound	Max bound	Multiperiod	Data Source	Latex syntax	Bibliography	Comment
Tmilk	DECISION	Milk inlet temperature	Mandatory	4	C	3	20	P,T,O(S)	./milkplant_data.csv	T_{milk}	@Florez-Orrego et al	Infeed temperature of the milk from transportation truck
Ta	CONTEXT	Environmental temperature	Optional	25	C	-273 C	1000	P,T,O(S)	./milkplant_data.csv	T_{a}	@Boztas et al	Environmental temperature used for the calculation of the PV efficiency (it is overwritten by the GIS information)
R	LITERATURE	Constant of the ideal gases	Private	8.314	kJ/kmolK	-	-	P,T,O(S)	./milkplant_data.csv	R_{gas}	@Ribeiro Domingos et al	Contant of the ideal gases
Lactose	MEASURED	Lactose fraction	Optional	2%	-	0%	5%	P,T,O(S)	./milkplant_data.csv	-	@Bhatnagar and Baibhav	Content of Lactose of the milk

Model sharing, ontologies and procedures

unit name	Minimum Capacity Multiplier	Maximum Capacity Multiplier
CheeseProduction	1	1

Reference sizes and upper/lower caps are important to establish the implementation range of the model.

CONNECTORS								
NAME (ALIAS)	TYPE	DIRECTION	VALUE	PHYSICAL UNIT	T [C]	P [BAR]	COMPOSITION [MOLE/MASS]	PHASE FRACTION [MASS]
MILK	MASS	IN	100	kg/h	25	1	MASSFRAC:FAT[0.03]&H2O[0.97]	-1 (L)
SALT	MASS	IN	1	kg/h	25	1	MASSFRAC:NACL[1]	-1 (S)
CHEESE	MASS	OUT	10	kg/h	25	1	MASSFRAC:FAT[0.3]&H2O[0.6]&NACL[0.1]	-1 (S)
WASTEWATER	MASS	OUT	91	kg/h	25	1	MASSFRAC:H2O[0.99]&FAT[0.01]	-1 (L)
LOWVOLTAGE	ELECTRICITY	IN	3	kW	-	-	-	-
STEAM	MASS	IN	50	kg/h	SAT	3	MASSFRAC:H2O[1]	1 (V+L)
CONDENSATE	MASS	OUT	50	kg/h	SAT	3	MASSFRAC:H2O[1]	0 (V+L)

Mass flows can go into or out from the sub-unit(s) of the model.

HEAT STREAMS								
name	Type	Tin [C]	Tout [C]	Hin [kW]	Hout [kW]	dTmin Contribution [C]	alpha [kW/m2K]	HeatCascade
pasteurization_heating	Cold	25.0	70.0	0.0	5.0	5	0.50	CUSTOM
pasteurization_quenching	Hot	70.0	25.0	5.0	0.0	5	0.50	CUSTOM
evaporation_preheating	Cold	25.0	90.0	0.0	2.0	5	0.50	DEFAULT
evaporation_latentheat	Cold	90.0	90.0	0.0	18	2.5	1.00	DEFAULT
condensation_latent	Hot	90.0	90.0	18.0	0	2.5	1.00	DEFAULT
condensation_subcooling	Hot	90.0	25.0	2.0	0.0	5	0.50	DEFAULT
cheese_curation	Cold	25.0	30.0	0.0	5.0	8	0.10	DEFAULT

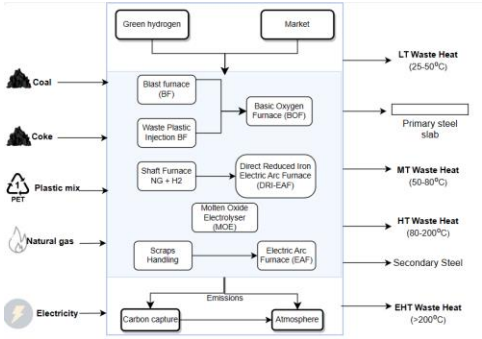
Pinch analysis (heat integration) requires the knowledge of the heat cascade and can include different heat cascades (constraints).

EQUIPMENTS				
name	Type	Parameters	Value	units
pump_milk	Pump	Power	2	kW
		Lifetime	20	years
		Date	2025	year
HEX_101	Heat Exchanger	Area	3000	m2
		Pressure	1	bar
		Lifetime	20	years
		Date	2025	year
Stirrer	Asset	Capital	75000	EUR
		Lifetime	20	years
		Date	2025	year

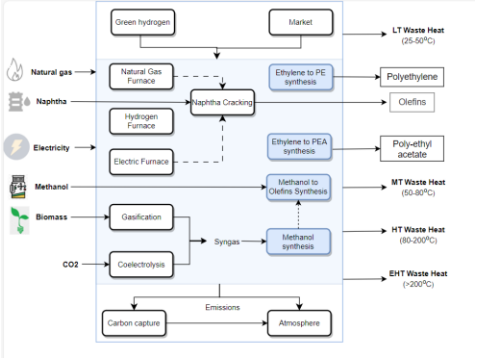
Reference sizes are important to adequately calculate the cost ranges and account for economies of scale.

AIDRES Database

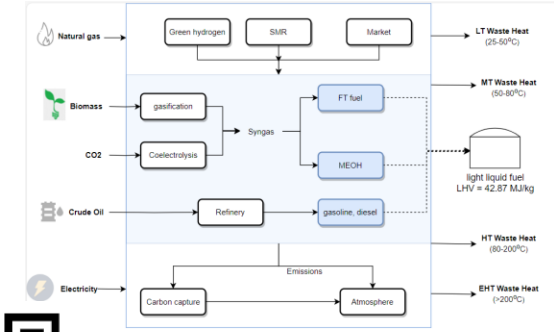
Steel sector



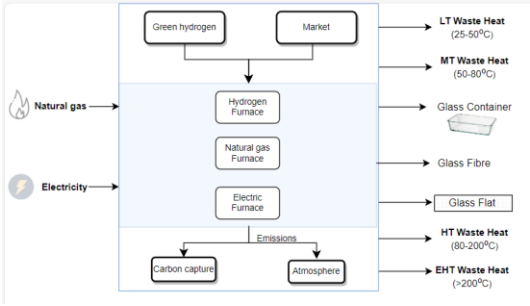
Chemical sector



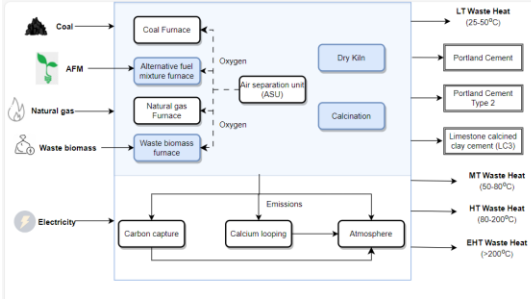
Refineries sector



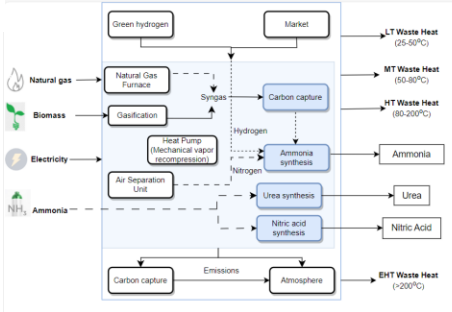
Glass sector



Cement sector



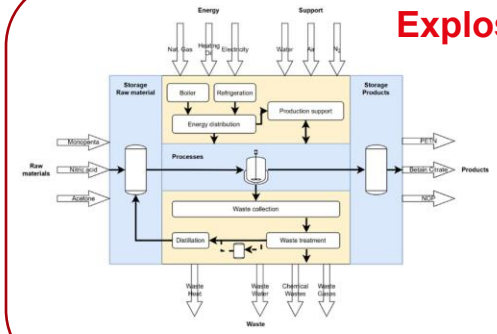
Fertilizer sector

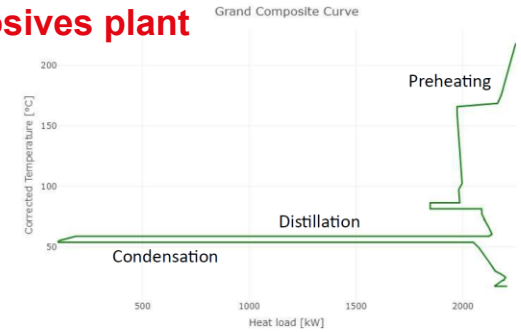


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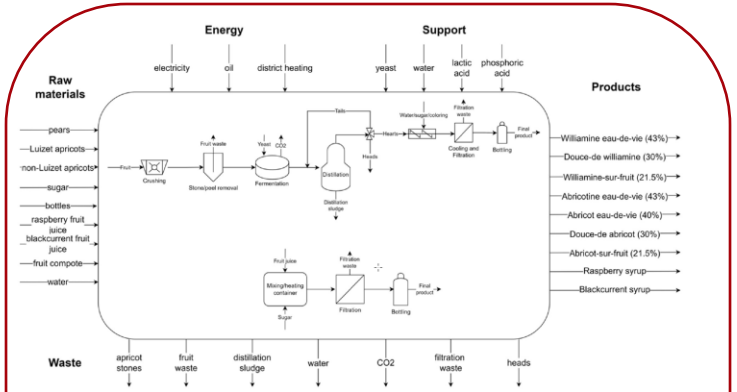
MAS audits database

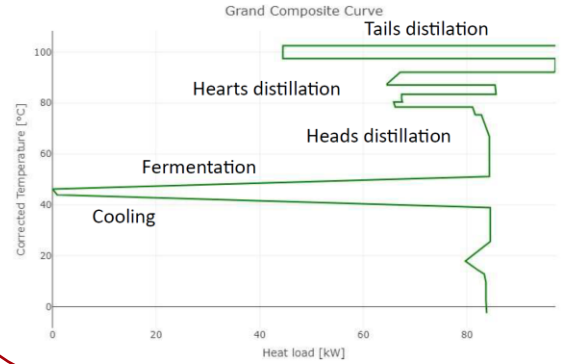
Explosives plant



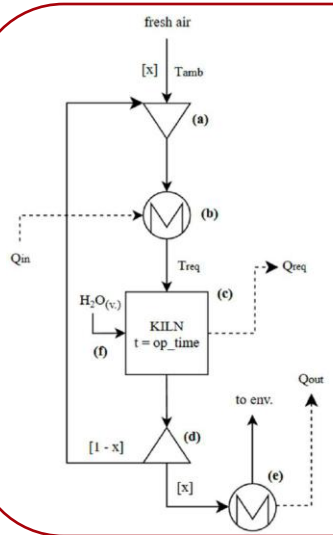


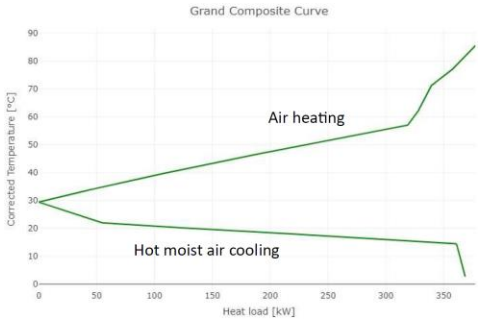
Distillery plant





Malting plant



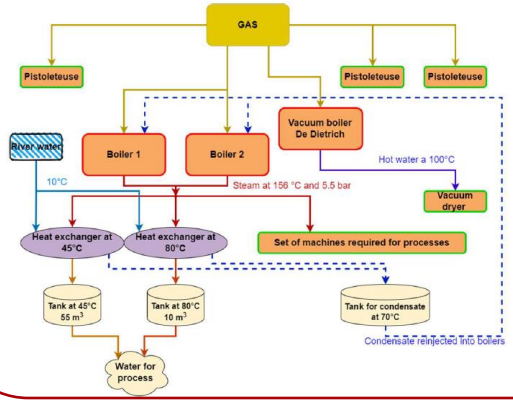


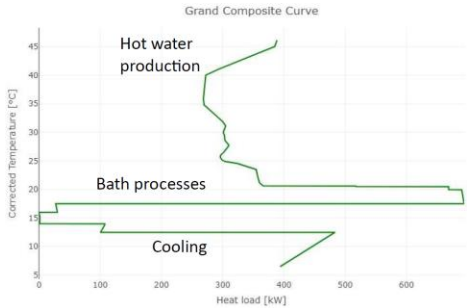
Florez-Orrego et al (2024) Swiss OFEN. https://pubdb.bfe.admin.ch/en/publication/download/11946

https://gitlab.epfl.ch/ipese/mas/masbook

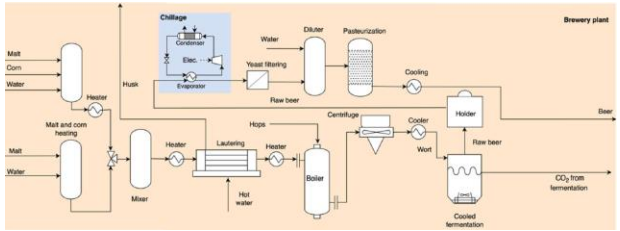
MAS audits database

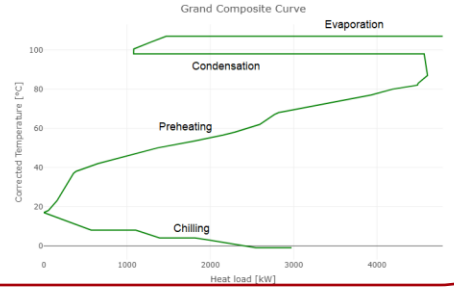
Tannery model



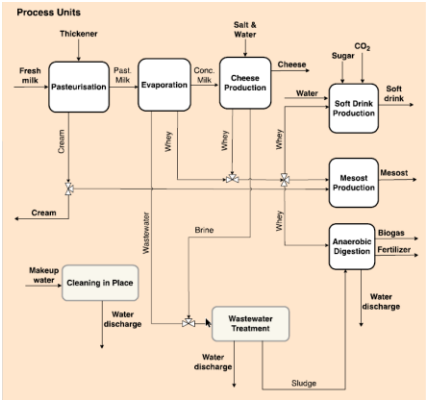


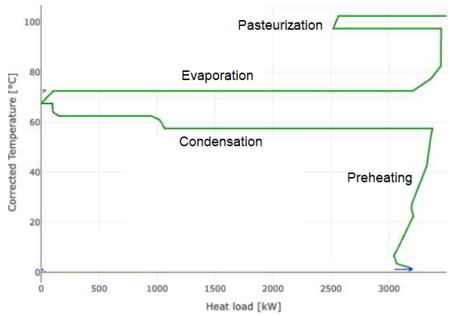
Brewery model





Dairy model

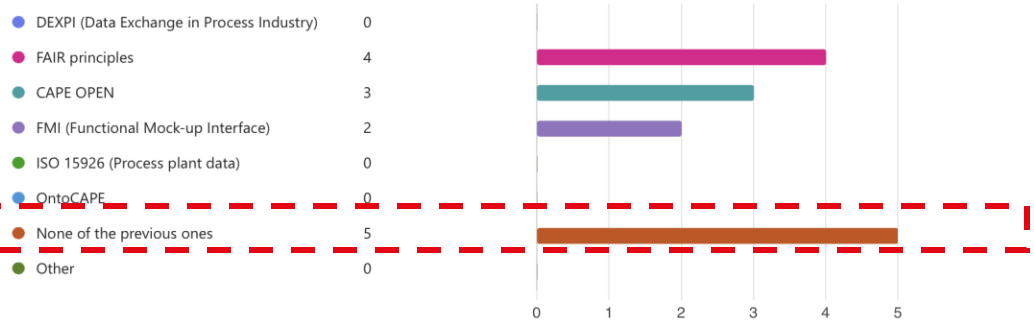




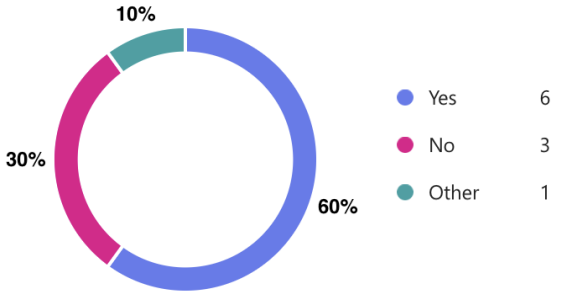
Florez-Orrego et al (2024) Swiss OFEN.
https://pubdb.bfe.admin.ch/en/publication/download/11946
https://gitlab.epfl.ch/ipese/mas/masbook

Formal *vs* simplified frameworks

27. Are you familiar with one or more of the following practices, standards and ontologies for formal models representa on?



Would you support developing a simplified, pragmatic approach specifically for Task XXIV to facilitate faster adoption?

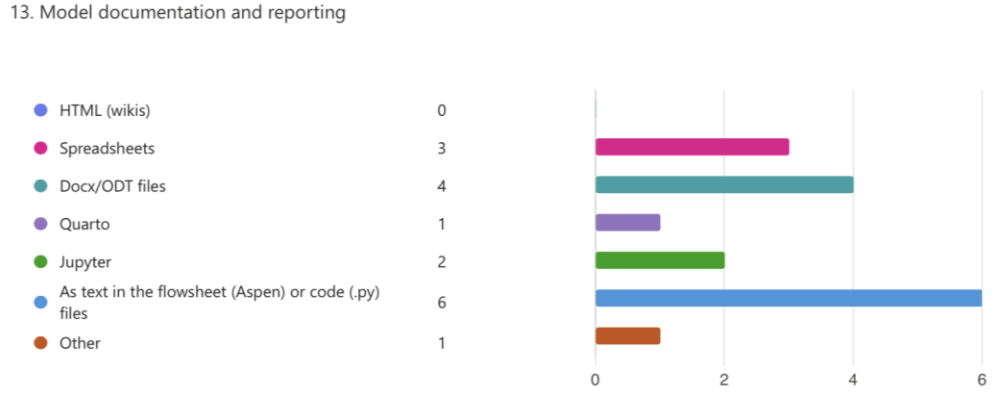


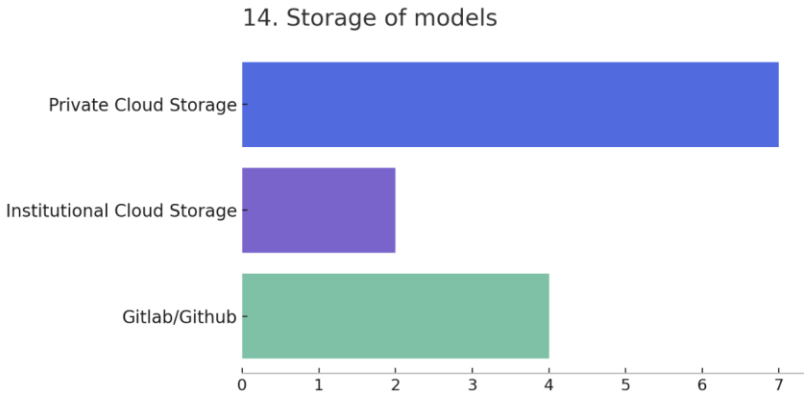


Overview of the survey results

Models inventory and documentation framework (database features and deployment)

Model tracking, documentation format and storage



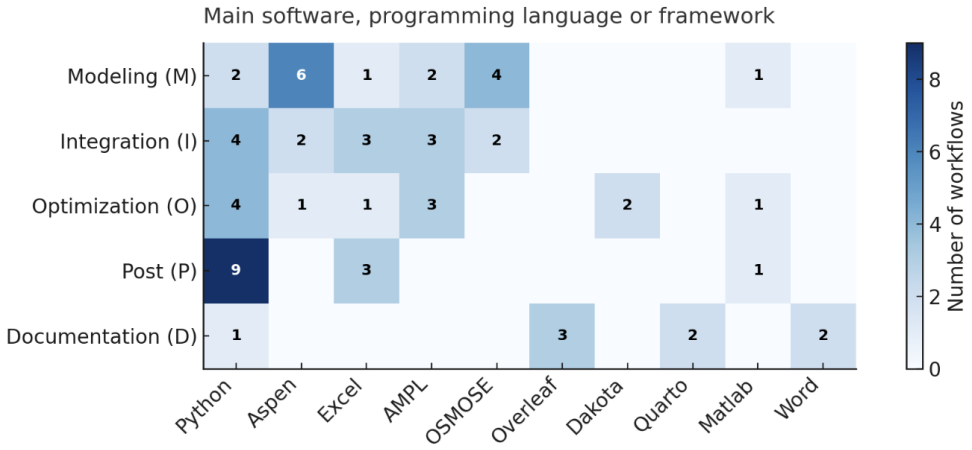


- **Aspen annotation**, **convenient** but highly **personalized, not** easily accessible, behind licensed software

- Models are maintained in an internal Gitlab, with each user managing their own documentation. Documentation is **standardized** through a Gitlab **Markdown Wiki (Who?) --> Workshop/Bilateral**

- A **centralized platform** could standardize model descriptions and **host the latest official version**, while external institutional repositories manage detailed **development, versioning, and history,** similar to how PyPI centralizes distribution while code lives on git-based repositories.

Set of tools used by the participants

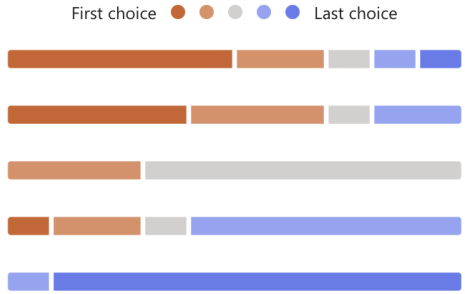


- **Most common workflow:** Aspen/OSMOSE for modeling + AMPL for integration/optimization + Python for post-processing.
- **Python is universal** for data handling, KPIs, visualization, and optimization wrappers.
- **Excel still widely used** for interfacing and calculator blocks.
- **Documentation split** between Overleaf/LaTeX/Quarto and Word-based reporting.

Paradigms for the modelling and optimization

Modelling methodology

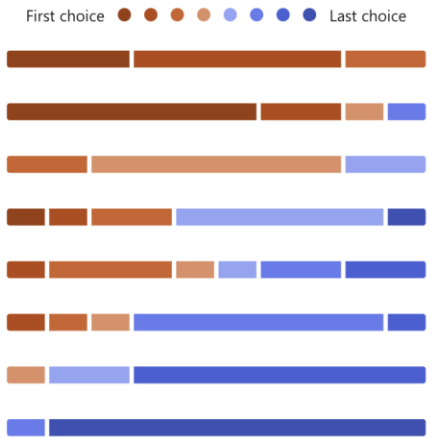
Rank	Options
1	Sequential modular (e.g. Aspen)
2	Equation oriented
3	Surrogate (data-driven)
4	Hybrid approach
5	Other(s)



- Define integration paths for flowsheet tools and surrogate models.

Optimization methodology

Rank	Options
1	Mixed Integer linear programming (MILP)
2	Linear programming (LP)
3	Non-linear programming (NLP)
4	Mixed integer non-linear programming (MINLP)
5	Integer programming (IP)
6	Metaheuristic programming (e.g. GA, SA, PSO)
7	Stochastic programming
8	Other



Feedbacks on the Model Sharing Template

METADATA

Model Name	DairyProcessing
Authors and Contributors	
Version	1.0
Confidentiality	Open
Creation Date	
References	
Grade	GRAY-BOX
Software	DWSIM
URL	
Last Update Date	
Keywords	Food
Reference Capacity (main product/feedstock flowrate)	10 kg/h Cheese

- Add **DOI, publication info**, **author contacts.**
- Include model status (maintained vs archived).
- Include list of supplementary files.

Feedbacks on the Model Sharing Template

CONNECTORS

EXTERNAL CONNECTORS							
Unique Identifier Number, UID	NAME (ALIAS)	TYPE	PHYSICAL UNIT	T [C]	P [BAR]	COMPOSITION [MOLE/MASS]	PHASE FRACTION [MASS]
MILK-FAT3	MILK	MASS	kg/h	25	1	MASSFRAC:FAT[0.03]&H2O[0.97]	-1 (L)
CHEESE-SOFT	CHEESE	MASS	kg/h	25	1	MASSFRAC:FAT[0.3]&H2O[0.6]&NACL[0.1]	-1 (S)
SALT-NACL	SALT	MASS	kg/h	25	1	MASSFRAC:NACL[1]	-1 (S)
WASTEWATER-SEWAGE1	WASTEWATER	MASS	kg/h	25	1	MASSFRAC:H2O[0.99]&FAT[0.01]	-1 (L)
TAP-WATER	WATER	MASS	kg/h	25	1	MASSFRAC:H2O[1]	-1 (L)
WOOD-DRY	WOOD	MASS	kg/h	25	1	MASSFRAC:C[0.47]&H[0.06]&O[0.45]&S[0]&ASH[0.02]	-1 (S)
NATURALGAS	CH4	MASS	kg/h	5	1	MASSFRAC:CH4[1]	-1 (V)
ELEC-50KV	MEDIUMVOLTAGE	ELECTRICITY	kW	-	-	-	-
ELEC-13KV	LOWVOLTAGE	ELECTRICITY	kW	-	-	-	-
FLUEGAS-M	CO2INFLUEGAS	EMISSIONS	kg/h	120	1	MOLEFRAC:CO2[0.15]&H2O[0.12]&N2[0.73]	-1 (V)
CO2-PURE	CO2	EMISSIONS	kg/h	25	1	MASSFRAC:CO2[1]	-1 (V)

Suggestions:

- Clearer handling of **phase state and phase fractions** (S/L/V separation).
- Including CO_2 impact and prices. ***It is subjective to the location and model users***

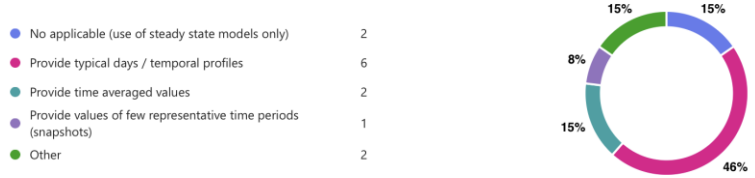
VARIABLES

VARIABLES											
Name	Type	Full name	User grade	Default Value / Formula	Physical unit	Min bound	Max bound	Multiperiod	Data Source	Latex syntax	Bibliography	Comment
Tmilk	DECISION	Milk inlet temperature	Mandatory	4	C	3	20	P,T,O(S)	./milkplant_data.csv	T_{milk}	@Florez-Orrego et al	Infeed temperature of the milk from transportation truck
Ta	CONTEXT	Environmental temperature	Optional	25	C	-273 C	1000	P,T,O(S)	./milkplant_data.csv	T_{a}	@Boztas et al	Environmental temperature used for the calculation of the PV efficiency (it is overwritten by the GIS information)
R	LITERATURE	Constant of the ideal gases	Private	8.314	kJ/kmolK	-	-	P,T,O(S)	./milkplant_data.csv	R_{gas}	@Ribeiro Domingos et al	Contant of the ideal gases
Lactose	MEASURED	Lactose fraction	Optional	2%	-	0%	5%	P,T,O(S)	./milkplant_data.csv	-	@Bhatnagar and Baibhav	Content of Lactose of the milk

Clarifications:

- **User grade:** It is defined to guide the users of the models to configure and adapt the models
- **Multiperiod:** For providing the temporal variations in a model ***(data inputting needs to be defined)***
- **Latex syntax:** Not necessary, can be removed
- **ReadtheDocs** about the template, variables and model

26. How would you handle models data sharing when they include batch, partial load, or dynamic aspects:



Feedbacks on the Model Sharing Template

unit name	Minimum Capacity Multiplier	Maximum Capacity Multiplier
CheeseProduction	1	1

CONNECTORS								
NAME (ALIAS)	TYPE	DIRECTION	VALUE	PHYSICAL UNIT	T [C]	P [BAR]	COMPOSITION [MOLE/MASS]	PHASE FRACTION [MASS]
MILK	MASS	IN	100	kg/h	25	1	MASSFRAC:FAT[0.03]&H2O[0.97]	-1 (L)
SALT	MASS	IN	1	kg/h	25	1	MASSFRAC:NACL[1]	-1 (S)
CHEESE	MASS	OUT	10	kg/h	25	1	MASSFRAC:FAT[0.3]&H2O[0.6]&NACL[0.1]	-1 (S)
WASTEWATER	MASS	OUT	91	kg/h	25	1	MASSFRAC:H2O[0.99]&FAT[0.01]	-1 (L)
LOWVOLTAGE	ELECTRICITY	IN	3	kW	-	-	-	-
STEAM	MASS	IN	50	kg/h	SAT	3	MASSFRAC:H2O[1]	1 (V+L)
CONDENSATE	MASS	OUT	50	kg/h	SAT	3	MASSFRAC:H2O[1]	0 (V+L)

HEAT STREAMS								
name	Type	Tin [C]	Tout [C]	Hin [kW]	Hout [kW]	dTmin Contribution [C]	alpha [kW/m2K]	HeatCascade
pasteurization_heating	Cold	25.0	70.0	0.0	5.0	5	0.50	CUSTOM
pasteurization_quenching	Hot	70.0	25.0	5.0	0.0	5	0.50	CUSTOM
evaporation_preheating	Cold	25.0	90.0	0.0	2.0	5	0.50	DEFAULT
evaporation_latentheat	Cold	90.0	90.0	0.0	18	2.5	1.00	DEFAULT
condensation_latent	Hot	90.0	90.0	18.0	0	2.5	1.00	DEFAULT
condensation_subcooling	Hot	90.0	25.0	2.0	0.0	5	0.50	DEFAULT
cheese_curation	Cold	25.0	30.0	0.0	5.0	8	0.10	DEFAULT

EQUIPMENTS				
name	Type	Parameters	Value	units
pump_milk	Pump	Power	2	kW
		Lifetime	20	years
		Date	2025	year
HEX_101	Heat Exchanger	Area	3000	m2
		Pressure	1	bar
		Lifetime	20	years
		Date	2025	year
Stirrer	Asset	Capital	75000	EUR
		Lifetime	20	years
		Date	2025	year

- Additional information on heat streams: *heat capacity, mass flows*
- It is important to provide also the tools to integrate the technologies in PA.
- How would a technology model look like in this case, with the proposed interface (**CC GCC --> screenshoting solution in the readme rmd of the models --> Postprocessing**)

- Add CAPEX **references**, equipment **subtype** type, and **material** of construction for all units.

Feedbacks on the Model Sharing Template

Q: Should there be multiple template variants tailored to different model categories and simulation approaches?

- **One template enough:** 4 task member
- **Special templates needed:** 3 task member



Hybrid approach proposed:

- Build **one general template:** Base (mass flow, production, energy conversion)
- Use a **hierarchical or layered structure**: Proposal of a tree --> **Three of objects**
- Add **category-specific extensions:** Optional detailed blocks depending on model type (*superstructure*)

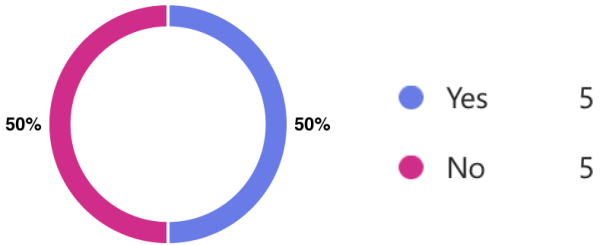
Summary of findings, and further development of ways of sharing

- The tools to be used for the shared database needs to be defined:
  - Excel is currently for **communication and standardizing the model-sharing** definitions, but not necessarily intended as the **final solution**
  - **Storage** of models, and **uploading** must be a **familiar and easy platform** for all the participants

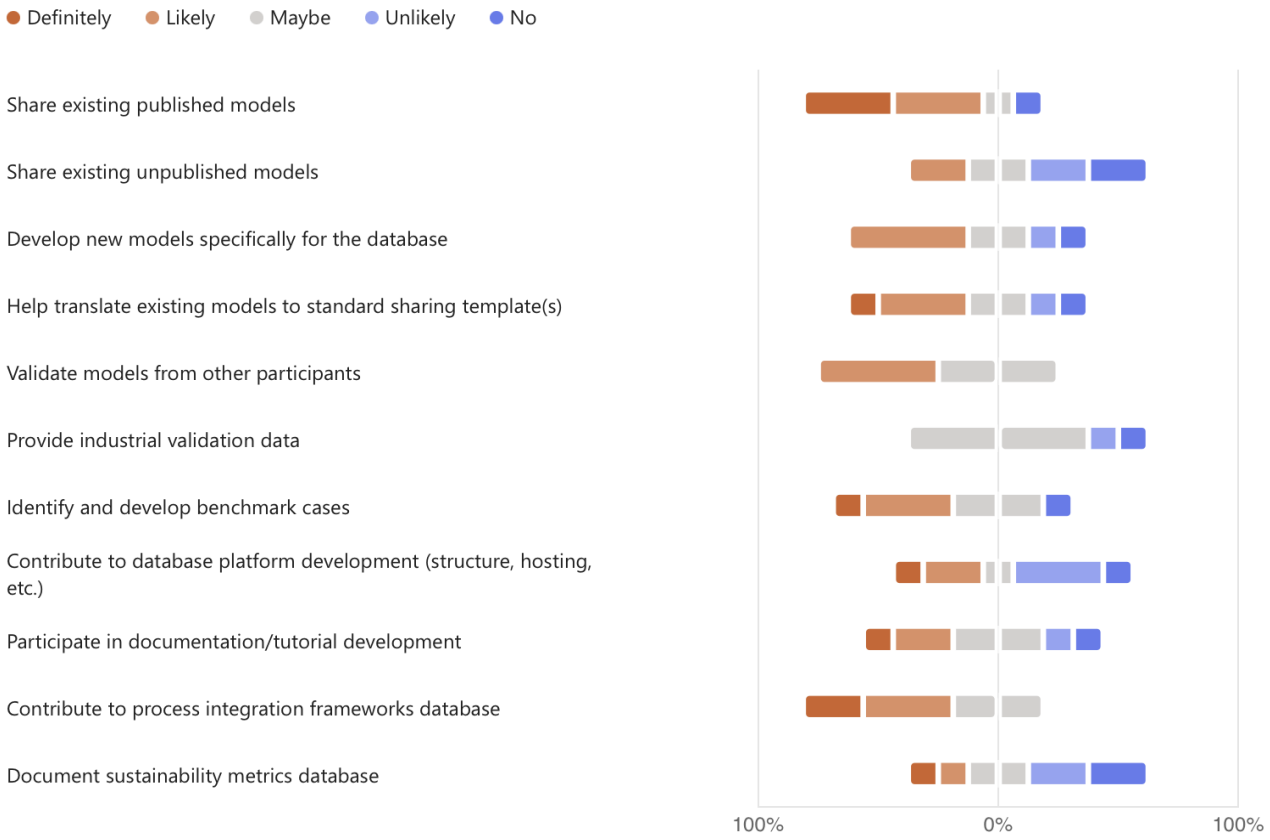
- Core template is broadly accepted, but feedback highlights the need to include:
  - Improved definition of connector quality: robust phase and composition definitions
  - **heat capacity and flowrate**
  - CAPEX references, equipment type, material of construction

- Several concerns noted on the applicability of the template for the broader options of mode
  - A **tree** for the **model definition** can be shared with the participants
  - **Model translation** trials can be started with the partners
  - In case of development of **special case templates**, discussion can be held with smaller groups who are **willing to contribute for the template.**

Q: Would your group or consortium be interested in developing an alternative template structure or formal model representation to be proposed to the Task XXIV participants for discussion?



Model sharing, ontologies and procedures



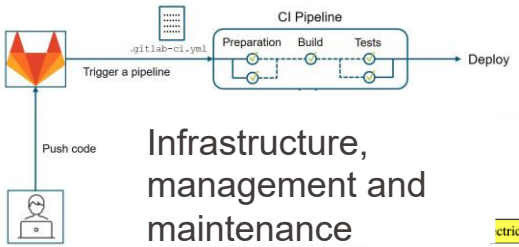
Deployment and maintenance challenge

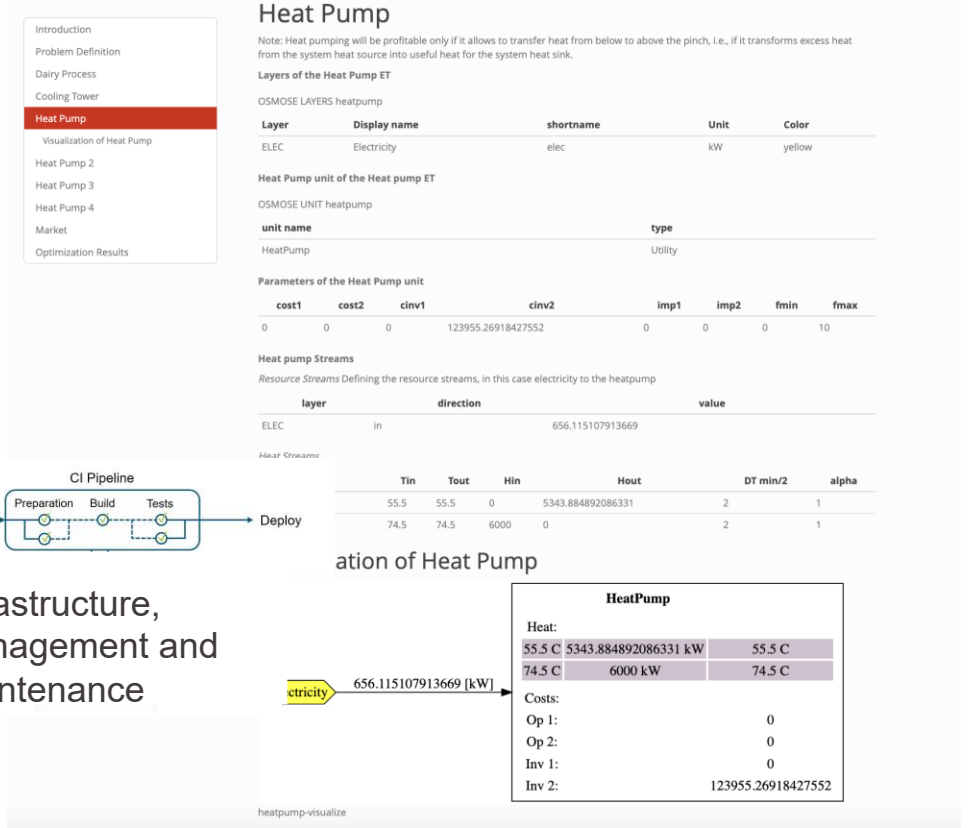
Syntax, terminology and specifications

Infrastructure, management and maintenance









MILP integration framework (OSMOSE example)

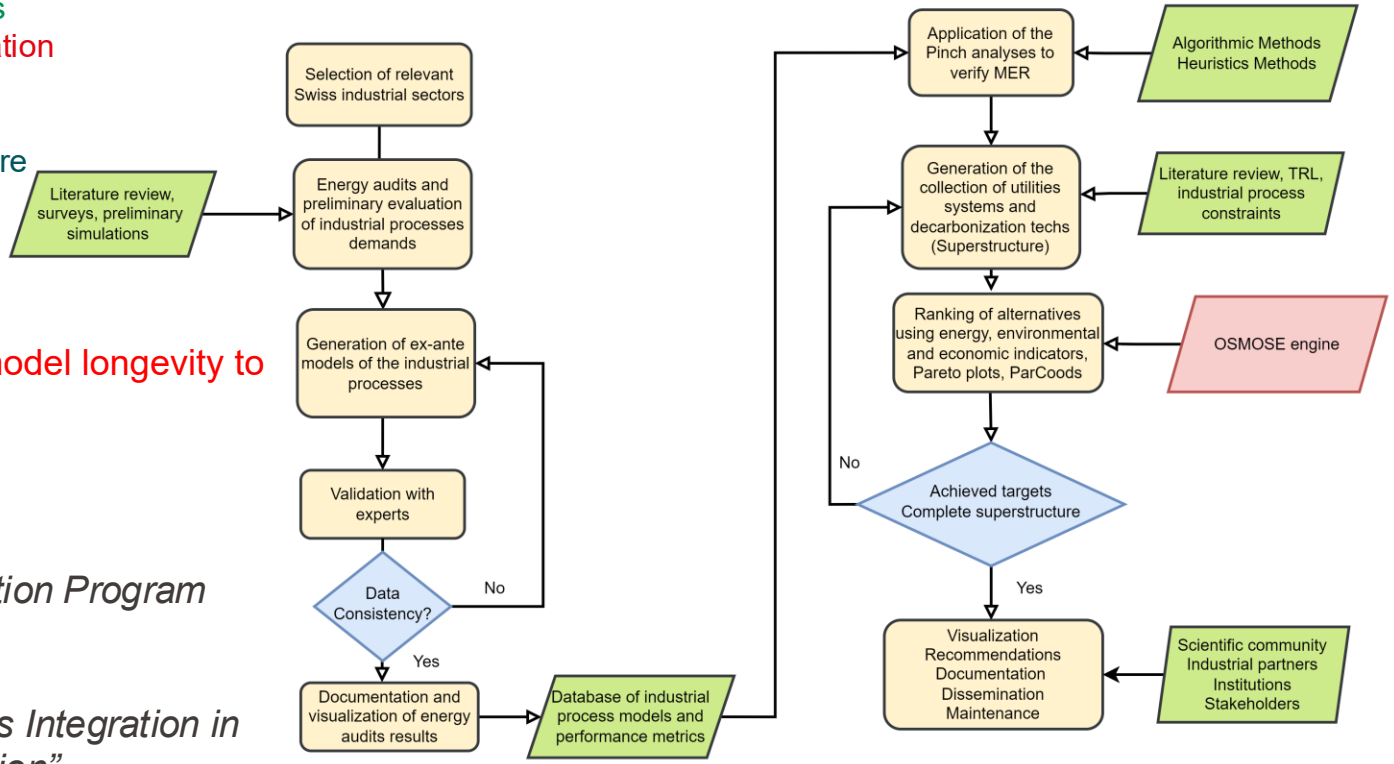
Support engineering decisions with flexible calculation and reporting tools.

Specialists surveys
Manufacturers information
Open literature
Case studies
Flowsheeting software

From project-lifetime model longevity to

IEA IETS
Technology Collaboration Program

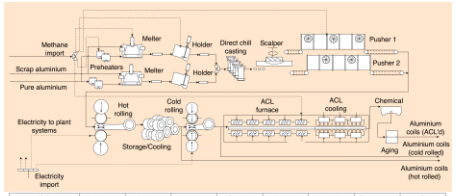
Task XXIV
"On the Role of Proces Integration in Industry Decarbonization"



Models of industrial processes and decarbonization technologies

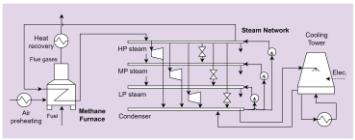
Speaker

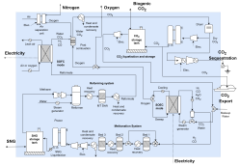
Industrial process DB

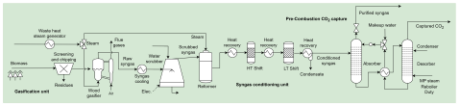


Unit	Preheater	Melter	Holder	Casting	Pusher	Hot rolling	Cold rolling	ACL
Energy load (kWh/tA)	30 - 40	350 - 500	30 - 40	180 - 200	190 - 250	50 - 70	40 - 60	100 - 150
Temperature level (°C)	150 - 250 °C	700 - 750 °C	700 - 750 °C	100 - 50° C	400 - 550 °C	300 - 450°C	50 - 150 °C	400 - 600 °C

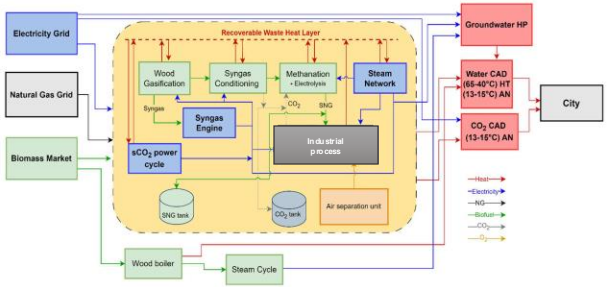
Utility systems and decarbonization technologies DB



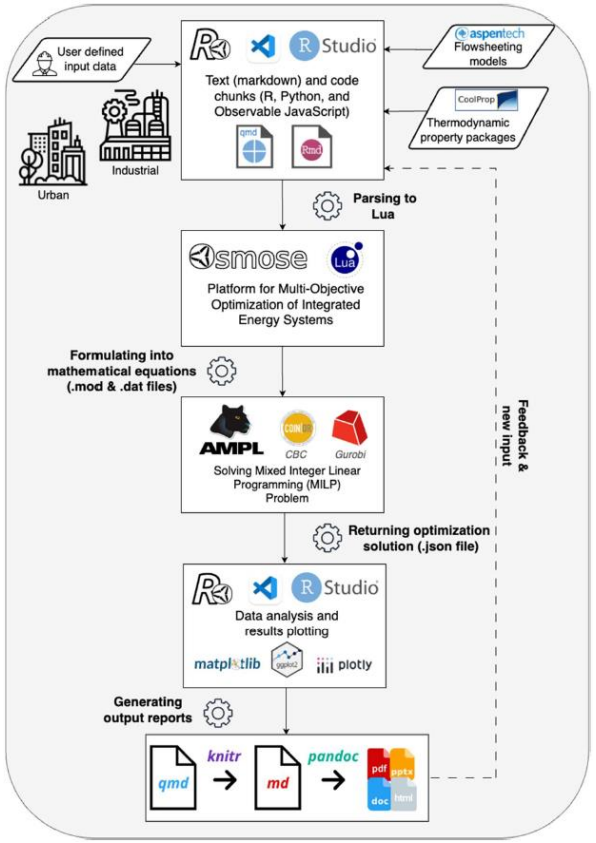




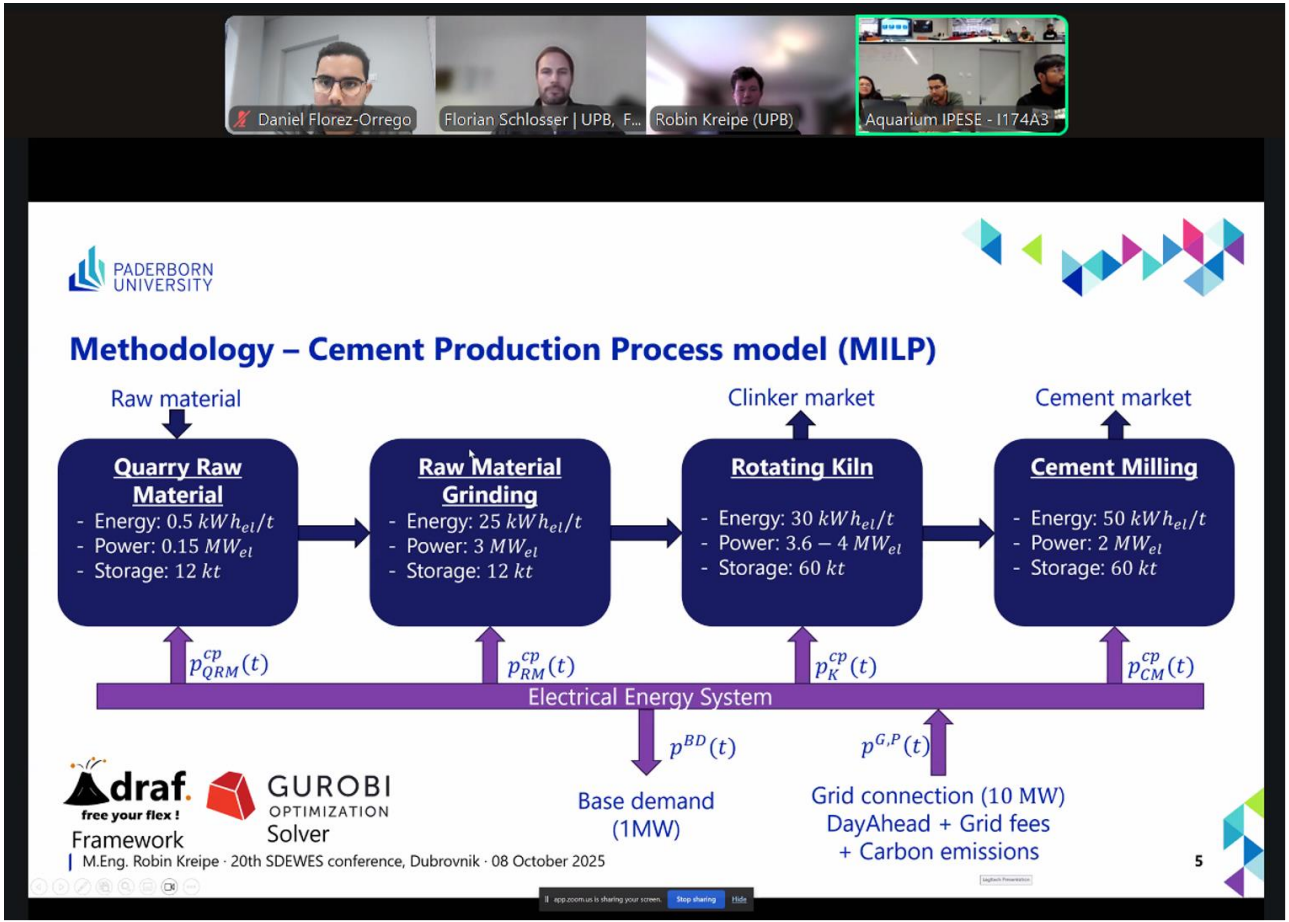
Integrated processs, technologies and energy systems



- Collect, **benchmark** and manage data: **flowsheeting**, industry **validation** & constraints
- **Inventory** and detailed description of process unit operations (process needs and interfaces)
- **Characterization** of mass, energy, information, emissions, costs flows.
- Propose energy efficiency improvement and **decarbonization** options
- Issue **recommendations** on decarbonization roadmaps



Other modeling frameworks



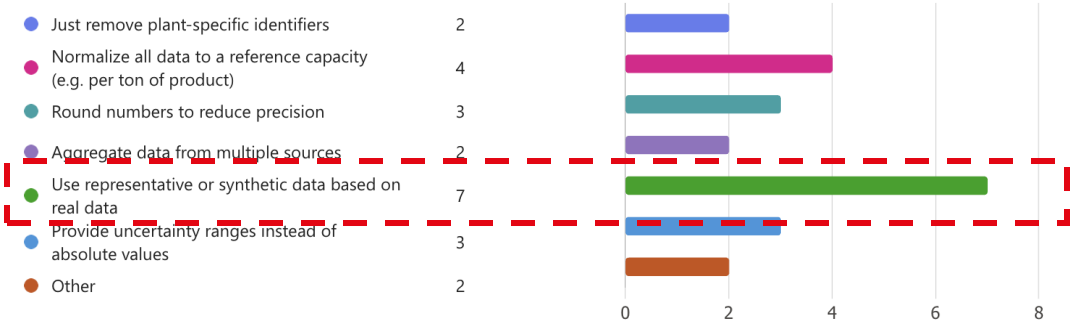


Overview of the survey results

Procedures, rules of the game, expectations, concerns and confidentiality issues

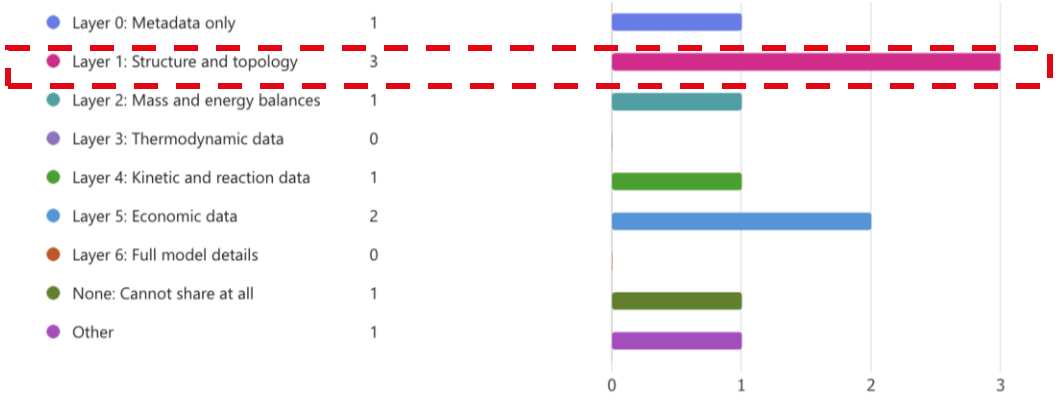
Industrial data confidentiality

Anonymization techniques



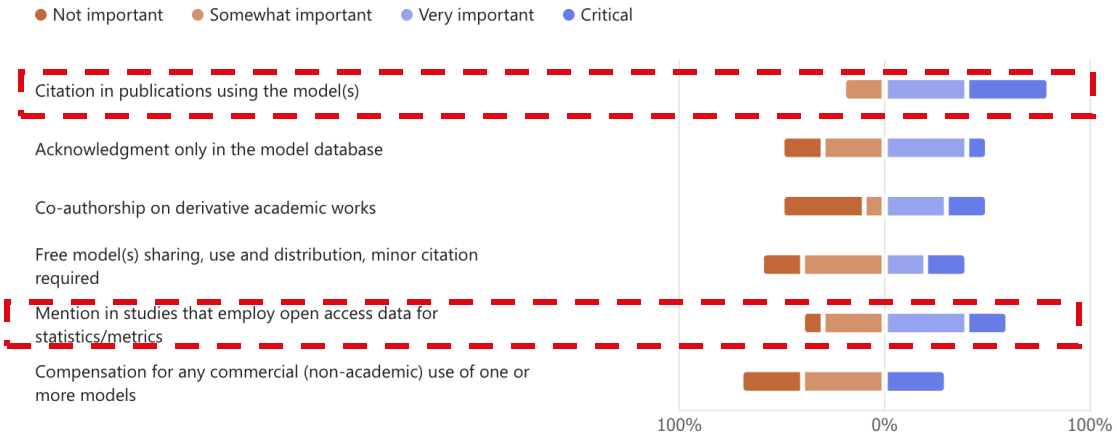
Synthetic data based on real data

Layer sharing for sensitive models



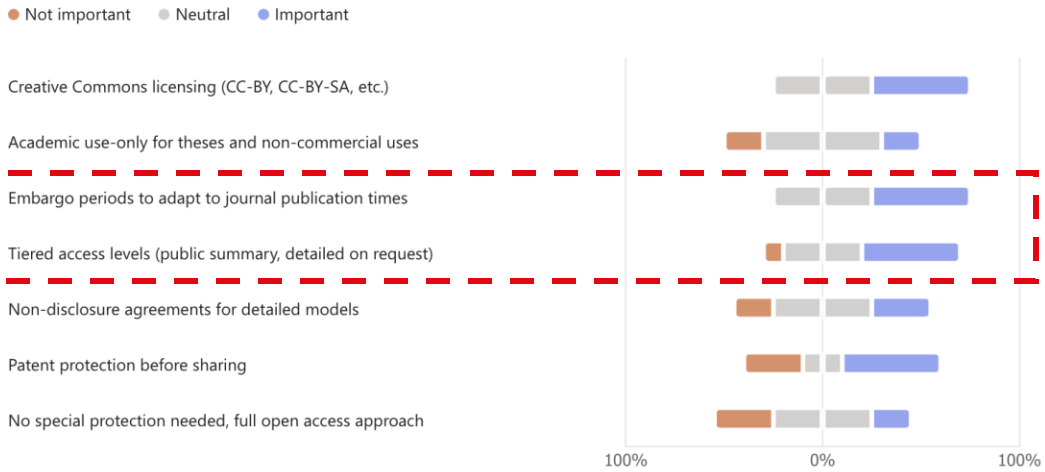
Structure and topology sharing for sensitive models

Recognition & Credit



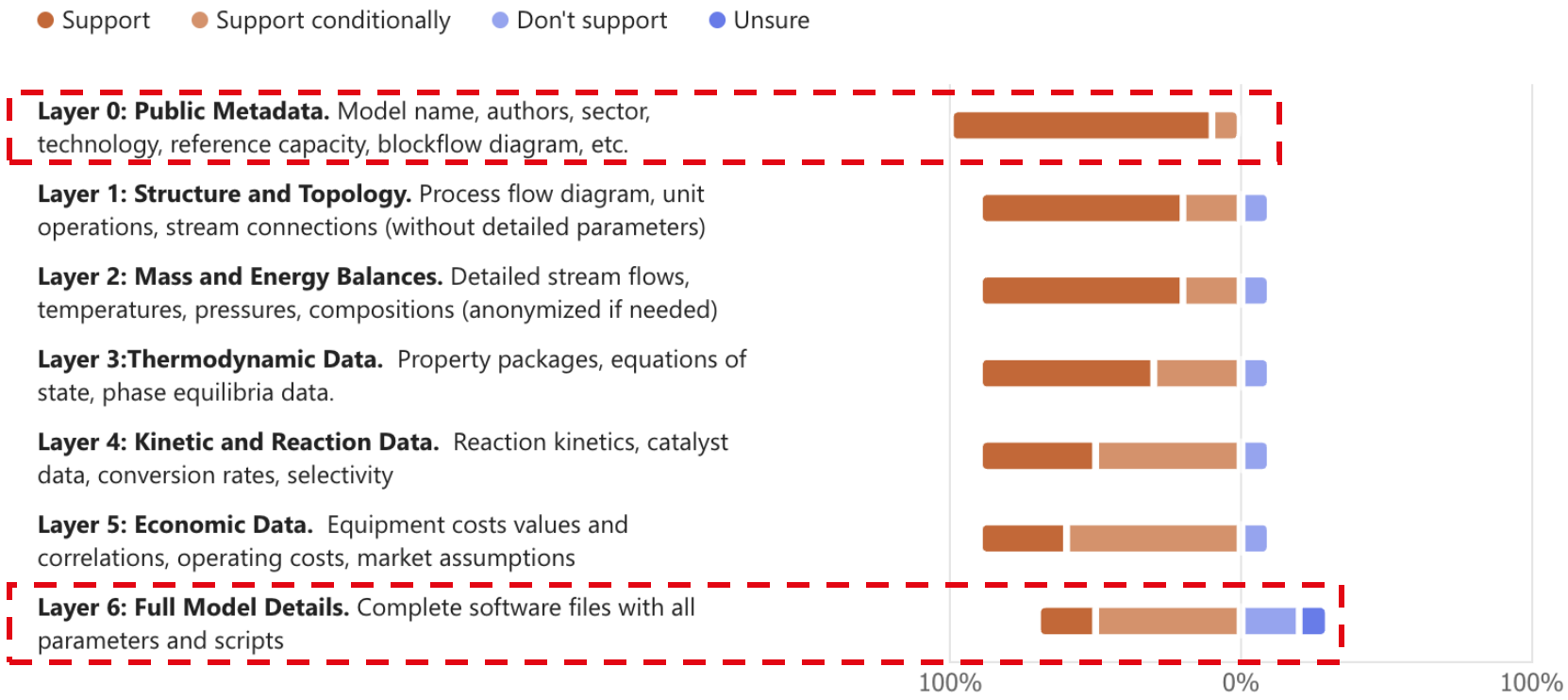
- "Recognition depends on the **level of model use** (e.g., is central to the research publication, entails substantial model adaptation, or is only ancillary use)"
- "Ideally, using the model leads to wanting to expand the model, and encourages **collaboration among the original developers** of the model and the person interested in it."
- "Since we are not allowed to share our developed models due to **Danish law**, therefore, the main collaboration means could be central to **research publication**."
- "I prefer **standardized guidelines."**
- "Fraunhofer Chile believes recognition should be commensurate with the **model's level of contribution to the research**. While standard citation suffices for auxiliary use, we expect **co-authorship or explicit acknowledgment** when the model is central to the findings or involves substantial adaptation. To streamline collaboration, we favor establishing **standardized guidelines** reflecting these tiers of usage from the outset, rather than negotiating protocols on a case-by-case basis."
- Recognition protocols should be negotiated on a **case-by-case basis**. Possibly a team could be assigned to review these processes, and author contacts should be given to facilitate this.

Intellectual property protection mechanisms



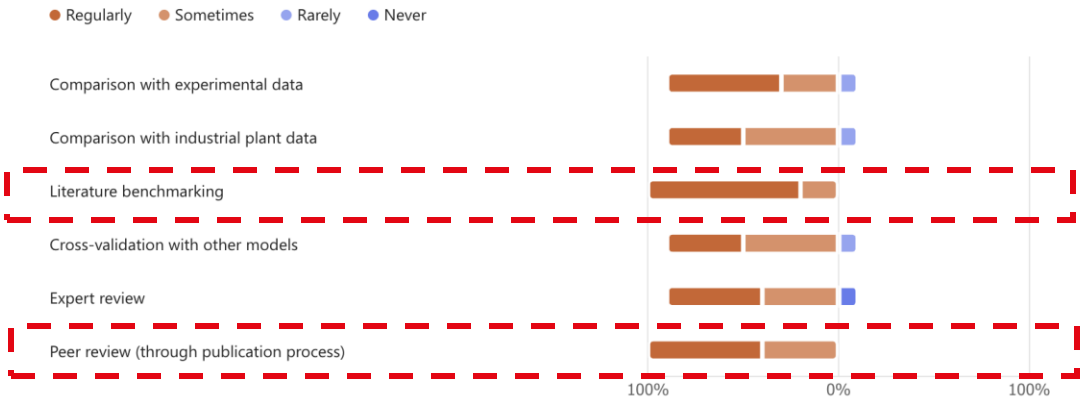
- "In Denmark, intellectual property created within public research institutions is governed by the **Danish Copyright Act**, including Section 59 on computer programs. Under this law, software, code, and models developed by employees as part of their official duties belong to the employer institution and ultimately to the Danish state. This applies to all models and code developed at SDU. Because of this legal framework, we are **not permitted to share code or executable models outside SDU without a formal agreement reviewed and approved by SDU's legal office.** Only high-level methodological descriptions or aggregated results can be shared openly. If more restrictive protection mechanisms are required, the only feasible option for us is a formal bilateral legal agreement that explicitly governs the terms of access and use. Without such an agreement, sharing our models is not legally allowed."
- "Regarding protection mechanisms, we require that **our proprietary models be distributed primarily as compiled binaries or functional mock-up units (FMUs)** (black-box approach) rather than open source code, to safeguard our intellectual property."

Multi-layered sharing approach



Model validation procedures

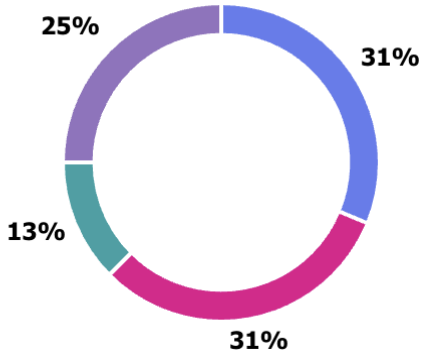
Procedures



- Literature benchmarking & Peer review
- Sub-teams creation to compare and validate specific models
- Open comparison among all Task XXIV participants

Control quality check

Create sub-teams to compare and validate specific models, then publish a comparison and validation...	5
Open comparison among all Task XXIV participants (requires a high-level evaluation system)	5
Define standard validation metrics for automated testing (more complex to implement)	2
Keep all the versions (validated and non-validated) regardless of the quality check result	4
Other	0

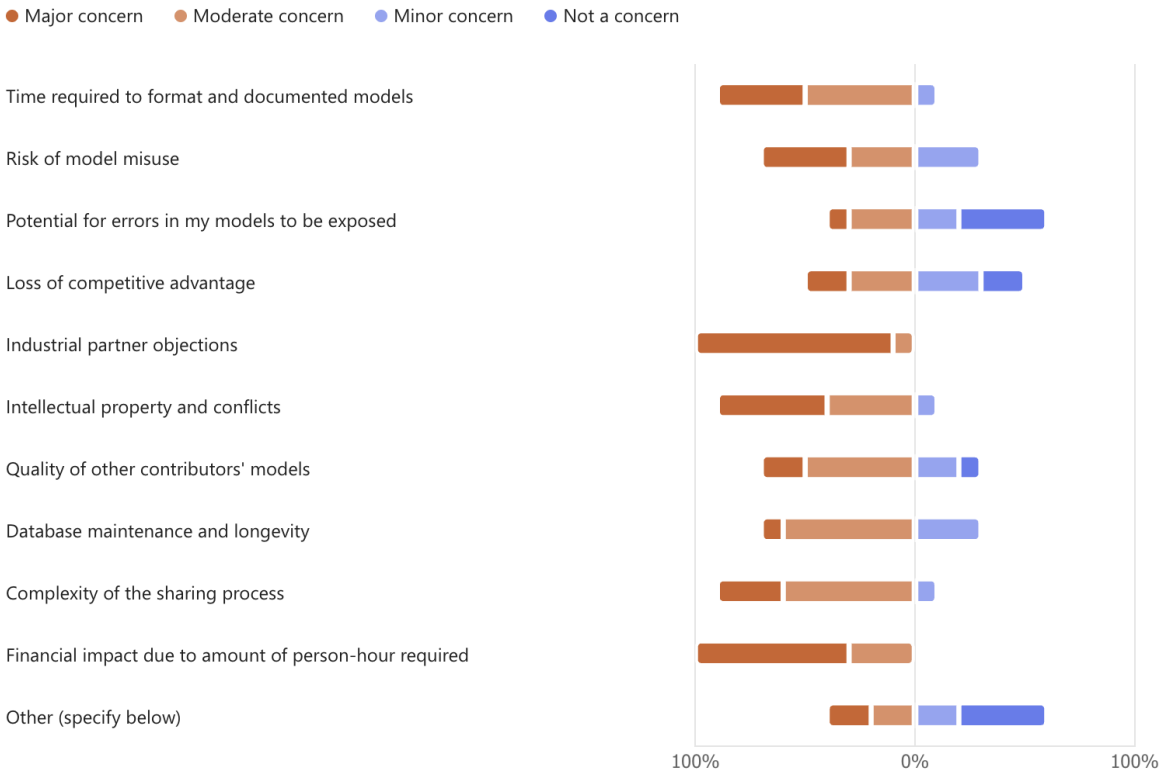


Expectations



- "To me, **"avoid reinventing the wheel**" is the main one."
- **"Get new projects / ideas."**
- "Discuss and define the definition, pros and cons, and barriers of shared databases of ex-ante models."
- "Sharing the models performed and validated to contribute to the Task overall goal."
- "we aim to **benchmark and validate our internal models** against international standards to ensure reliability for industrial applications. Our participation is driven by the goal of reducing technical uncertainty through standardization."

Concerns & Barriers



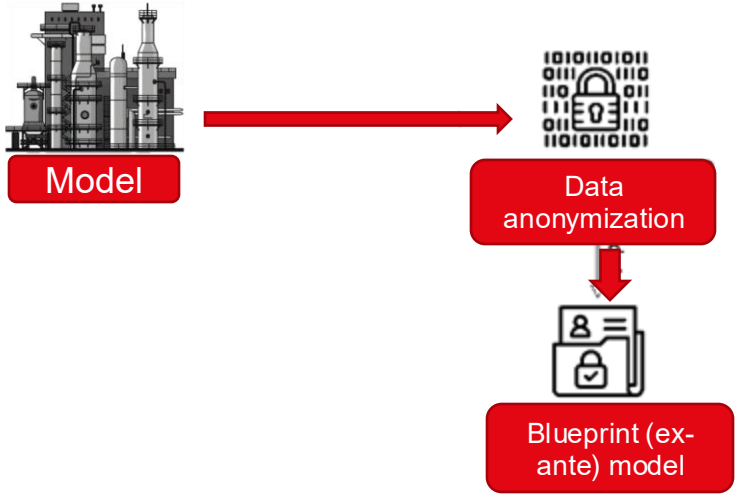
- **Time dedication**
- It will take **time** to get to know and use the database of models, and it needs a critical mass of models to be useful.
- **licensing** issues for commercial software.
- Participants have no **common understanding of legal barriers** to sharing ex-ante models.
- Going from Aspen plus to the template not being very straight forward (**time** consuming)

Confidentiality considerations

- Pseudonymization and anonymization

Achieved through generalization:

- Thermodynamics
- Obfuscation
- Aggregation



Primary recommendation for enabling, establishing and maintaining the evolving ex-ante model database

- "Define the **standard model structure template** and promote more rigorous peer review"
- **"Starting right away**, populating the database soon! Iterate and improve on the sharing framework as the task advances. **By doing it already, we can see what works and what does not."**
- "nothing, it's on good tracks. Main concern is the **time** needed to be dedicated to this project"
- **"Standardization"**
- "Knowledge sharing should be the main focus"
- "Need more information and more 1 to 1 discussions for further understanding"
- "The primary recommendation is to establish a **centralized framework**, an open-access platform modeled on research data sharing."
- "we emphasizes the critical need for **standardized documentation** and rich metadata for every model entry"
- "Provide a relevant and helping framework to access other academic work. Our goal is to tackle large scale and complex problems. **Why should many of us start from the beginning all the time and not having a foundation to build on?"**



Short presentations

6 min. Per participant country

Short presentations

- Participant countries describe their contributions and explain their thoughts on the following topics:
  - List of model(s) used as starting point. Is the 'simplified' models template compatible with the participants modelling approach?
  - Conditions to share regarding:
    - confidentiality
    - anonymization
    - citation
    - levels of model sharing



Break

5 min.



Break out session

15 min. discussion.
1 min. conclusions

Break out session dynamic

One moderator from the Task XXIV in each online room (4 rooms) discuss on the following topics:

- The 'simplified' data template and starting point (7 min.)
- The agreement regarding (7 min.):
  - Confidentiality
  - Anonymization,
  - Citation
  - Levels of model sharing

Moderators can help clarifying details of the models' template structure and summarize the main points on the "rules of the game" to be presented in 1 min. to the plenum.



Annual report contribution

Task XXIV Annual Report

- Min. 5 and max. 10 pages (e.g. editable slides, docs or LaTeX formats, although PDF files are also possible):

  i. brief description of your country's consortium (experience, research focus, other Tasks collaboration, etc.),

  ii. list of conferences and other events attended, as well as technical and journal publications that are related to the objectives of the Task XXIV,

  iii. exemplary diagram(s) or snapshots that briefly illustrate your models structure; any raw links to known/used model repositories (own or external) are also relevant,

  iv. simplified descriptive workflow and illustrative screenshots of your optimization framework(s) (in-house or proprietary software),

  v. examples of output reports, documentation excerpts, and other relevant content for the Annual Report of the Task XXIV



Next steps

Start right away!

- Annual report Task XXIV will define the "rules of the game".

- Subject to time availability, each partner can start with one (or more) shareable models to identify the 'simplified' interfaces that their own models can offer to the rest of the participants:

  - Metadata (authorship, software, level of sharing, etc.)
  - Mass, energy and emissions (and information) flows ("ports").
  - Documented (referenced) parameters and assumptions,
  - Description of main decision variables (input parameters) and derived (calculated) variables.
  - Simplified block flow diagram
  - Supplementary material (flowsheeting, publications, other files)

Models of industrial processes and decarbonization technologies

- Dairy
- Brewery
- Explosives
- Malting
- Tannery
- Distillery
- Aluminium
- Sugar
- Olefins
- Watchmaking
- Malting
- Fertilizers
- Cement
- Steel
- Glass
- Refinery
- Methanol
- Dimethyl ether
- Hydrogen
- Biomethane
- Hydro-Veg Oil
- Lime
- Biogas
- Waste upgrading
- Fischer Tropsch
- Alcohol to jet
- Biodiesel
- Ethanol 1&2G
- Hydrothermal gasification
- Data centers
- Urban systems

- Oxycombustion
- Conventional furnaces (G,S,L)
- Electrical furnaces
- Plasma furnaces
- Methane pyrolysis
- Thermal power cycles
  - ICE
  - GT
  - ST
  - ORC
- Polygeneration
- Steam network
- Cooling systems (air, water)
- Air separation (Cryo., VPSA, adsorb.)
- Refrigerator
- Electrolyzers

- Fuel cells (alkaline, PEM, SOC).
- Heat pumping (HP, MVR)
- PV panel
- Wind turbines
- Biogas purification
- CO_2 capture solvents
- CO_2 capture membranes
- CO_2 capture adsorption
- CO_2 mineralization
- CO_2 transportation
- CO_2 injection
- Gasification (syngas)
- Reforming and rWGS
- Methanation

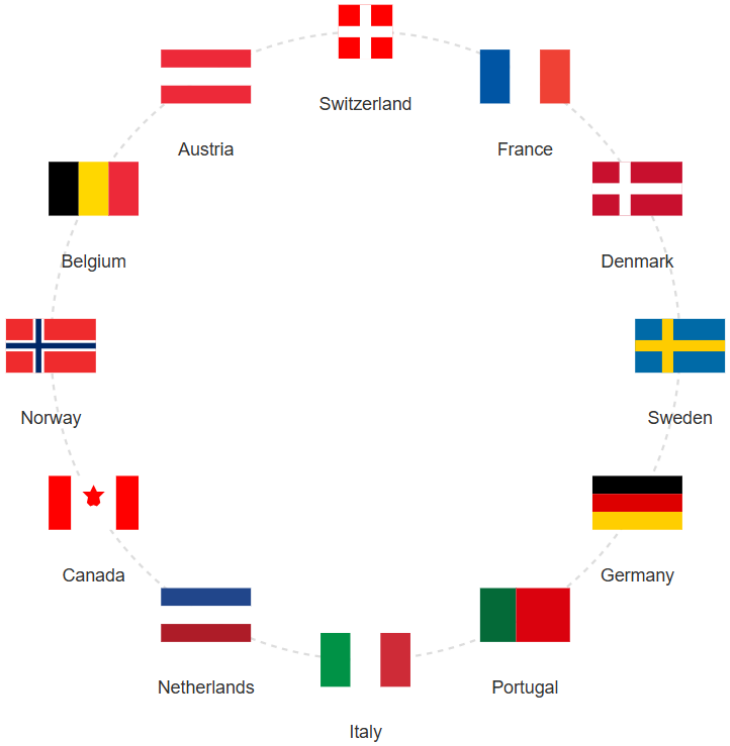


Task XXIV
Process integration for industry decarbonization

Subtask 1 Enabling a shared database of ex-ante models of industrial processes, decarbonization technologies and energy systems featuring sustainability metrics

https://iea-industry.org/tasks/process-integration-for-industry-decarbonization/

International participants (confirmed or invited)



Observers



- Pullah Bhatnagar
- Vibhu Baibhav
- Meire Ribeiro Domingos
- Assaad Zoughaib
- Benjamin Zühlsdorf
- Sanduni Pathiraja
- Roger Padullés I Solé
- Henrik Thunman
- Alexander Mitsos
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- Guillaume Boissonnet
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- Andrea Ramirez
- Etienne Bernier
- Serge Bedard
- Judit Fortet Casabella
- Thomas Adams II
- Muhammad Salman
- Du Wen
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- Zheng Grace Ma
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- Manuele Margni
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- Pedro Mendes
- Sara Russo Garrido
- Henrique Matos
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- Pedro Nunes
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- Nasrim Kermani
- Lijun Wu